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Executive Summary

Credit default management has become a critical concern for lending institutions as consumer borrowing volumes grow and portfolio risk becomes increasingly difficult to assess through traditional scorecard approaches. This study was undertaken to explore whether a data-driven, machine learning-based framework could provide a more accurate, timely, and actionable basis for credit risk decisions in a retail lending context.

The analysis is based on a dataset of 45,000 loan applicants, cleaned to 44,541 usable records. The dataset spans 14 variables covering borrower demographics, financial characteristics, loan terms, credit history, and repayment outcomes. The target variable — loan default (loan_status = 1) — affects 9,942 borrowers, representing an overall default rate of 22.32%.

Twelve formal hypothesis tests were conducted to identify which factors are statistically associated with default. Of these, nine returned significant results at the 5% level. The most powerful finding: interest rate ($t = 74.54$, $p < 0.001$), loan-to-income ratio ($t = 88.28$, $p < 0.001$), income ($t = -51.59$, $p < 0.001$), and prior default history ($\chi^2 = 13{,}173.5$, $p < 0.001$) are the dominant default drivers.

Three machine learning models were built to predict individual-level default risk: Logistic Regression, Decision Tree, and Random Forest. Random Forest delivered the best overall performance — ROC-AUC of 0.973, accuracy of 92.84% — followed by the Decision Tree (AUC = 0.960) and Logistic Regression (AUC = 0.947). Prior default history (importance = 0.345), interest rate (0.140), and the engineered debt-to-income ratio (0.104) are the three most important predictors.

The key finding of this study is that loan default in this portfolio is a financial affordability problem, not a demographic one. Lenders who focus their risk assessment on measurable indicators of financial strain — the interest rate charged, the proportion of income committed to repayment, housing stability, and employment history — will produce more accurate and equitable credit decisions.

Dimension	Key Result
Dataset	44,541 loan applicants; 22.32% default rate; 14 variables
Hypothesis Testing	9 of 12 hypotheses rejected; key drivers: interest rate, income, loan-to-income ratio, home ownership, prior default history
Best ML Model	Random Forest: ROC-AUC = 0.973, Accuracy =

Dimension	Key Result
	92.84%
Top Feature	Prior default history (importance = 0.345); interest rate second (0.140)
Risk Tiers	High: 29.7% | Medium: 11.0% | Low: 59.3% of test applicants
Survival Analysis	C-index ≈ 0.927; interest rate HR = 1.375; employment experience HR = 0.303
Not Significant	Credit score, education level, and gender show no significant association with default

Table ES.1: Summary of key findings across all analytical stages.

Chapter 1. Introduction

In recent years, the retail lending landscape has undergone a significant transformation, particularly in the domain of unsecured consumer credit. This evolution is driven by a complex interplay of socio-economic, technological, and behavioural factors that collectively reshape how lenders assess borrower risk and how consumers engage with credit products. Unlike credit markets in earlier decades, where the primary tools were manual underwriting and aggregated statistical tables, modern lending increasingly relies on large-scale data collection and automated decision systems.

The retail loan market has emerged as a pivotal component of the broader financial ecosystem. Market studies and regulatory analyses suggest that the growth trajectory of unsecured consumer credit — personal loans, debt consolidation facilities, and medical financing — continues to outpace more traditional secured lending products. Accurate prediction of loan default is thereby essential for a diverse array of stakeholders, encompassing commercial banks, fintech lenders, credit bureaus, regulators, and individual borrowers. Each entity within this ecosystem relies on precise credit risk assessments, directly or indirectly impacting their decision-making processes regarding profitability, regulatory capital, customer acceptance, and market competitiveness.

The Problem with Lending Blind

The retail lending industry is built on a single operational challenge: deciding, quickly and at scale, whether a given borrower is likely to repay a loan. Get it wrong in one direction and the institution absorbs a loss; get it wrong in the other and a creditworthy borrower is denied access to capital. Historically, this decision has been made through a combination of credit scores, income verification, and the intuition of experienced underwriters. These methods work up to a point — but they share a fundamental limitation. They are designed to measure risk at a single point in time, using a relatively small number of pre-defined inputs, and they do not improve as data accumulates. In a lending environment where millions of applications are processed each year and the cost of a missed default can run into thousands of dollars per case, this is a significant gap.

In the dataset analysed in this study — 45,000 retail loan applications — the default rate stands at 22.32%. For context, this means more than one in five borrowers did not repay their loan. At average loan amounts of approximately USD 9,500–10,800 depending on the borrower group, a portfolio of this size operating at a 22% default rate is sustaining losses that would materially compress margins and require substantial provisioning.

Why Standard Approaches Fall Short

Most lending institutions rely on three primary tools to manage credit risk: the credit score, the debt-to-income threshold, and the loan-to-value ratio (for secured lending). These metrics are useful but incomplete. Credit scores are backward-looking aggregates — they summarise a borrower's historical behaviour but do not capture current financial stress or the specific loan structure being applied for. In this study's dataset, credit scores are virtually identical between defaulters and non-defaulters (mean of 631.85 versus 632.70), confirming that the score alone is not discriminating between these two groups in any operationally meaningful way.

Standard income verification catches obvious cases of overextension but misses the interaction effects that matter in practice: a borrower with a high income and a very high loan amount may be more at risk than a lower-income borrower with a proportionally smaller loan. The debt-to-income ratio — which this study uses as an engineered feature — captures this interaction directly and proves to be the third most important predictor in the Random Forest model (importance = 0.104), ahead of credit score (0.043).

There is also a time dimension to default that static models ignore entirely. Survival analysis, borrowed from biomedical research, provides a framework for modelling this temporal dimension. Applied to the current dataset, the Cox Proportional Hazards model reveals that high-rate borrowers and renters not only default more frequently, they default earlier — a finding with direct implications for provisioning, early warning monitoring, and product design.

What Data Can Tell Us

The dataset used in this study contains 14 variables covering borrower age, gender, education, income, employment experience, housing status, loan amount, loan purpose, interest rate, loan-to-income ratio, credit history length, credit score, prior default history, and the binary outcome of whether the loan was repaid. Machine learning models are particularly well-suited to this kind of multi-variable, interaction-heavy prediction problem. Rather than assuming that each predictor has a fixed, linear relationship with the outcome, ensemble methods like Random Forest allow the model to discover which combinations of features — high interest rate plus high debt-to-income ratio plus rental housing, for example — are most predictive of default.

The results demonstrate that this approach works extremely well here: the Random Forest model achieves a ROC-AUC of 0.973 on the holdout test set, compared to 0.947 for Logistic Regression, with the difference attributable largely to the ensemble method's ability to capture non-linear interactions.

About This Study

This dissertation presents a credit risk prediction framework built on a dataset of 45,000 retail loan applicants. After cleaning — removal of implausible ages, extreme income outliers, and logical inconsistencies — 44,541 records were retained for analysis. The study works through the problem in stages. The raw data was cleaned and a derived debt-to-income ratio feature was engineered. The analysis then proceeded through exploratory investigation, twelve formal hypothesis tests, three machine learning models with risk tiering, and a Cox survival model.

The central question this study attempts to answer is not simply 'who will default?' — any reasonable model can generate a probability. The more important questions are: what is actually driving default in this portfolio, is it the loan structure, the borrower's financial position, or something else? And which of those drivers are measurable, actionable, and predictively stable enough to build a credit policy around?

Structure of the Dissertation

The remainder of this dissertation is organised as follows. Section 2 sets out the specific objectives of the study. Section 3 reviews the academic literature on credit risk prediction, machine learning in lending, and survival analysis in financial contexts. Section 4 states the twelve research hypotheses formally tested in this study. Section 5 describes the research methodology — the dataset, cleaning process, feature engineering, and analytical approach. Section 6 presents the full data analysis, including exploratory findings, hypothesis test results, model performance metrics, and survival analysis output. Section 7 discusses the key findings, draws conclusions, and proposes actionable recommendations for credit risk teams. Section 8 acknowledges the limitations of the study and outlines directions for future research.

Chapter 2 - Objective of the Study

1. **To identify the most important features affecting loan default probability.**
   The study employs feature importance and selection methods to assess the influence of various borrower-specific and loan-level attributes — such as interest rate, income, home ownership status, loan intent, and employment experience — on default outcomes.
2. **To predict loan default using machine learning techniques to overcome the limitations of traditional methods.**
   This entails the evaluation of multiple classification models including Logistic Regression, Decision Tree, and Random Forest to determine which algorithms offer the best predictive performance when applied to real-world lending data, assessed via ROC-AUC, recall, and F1-score.
3. **To segment borrowers into actionable risk tiers for credit policy use.**
   The research aims not only to build an accurate model but also to ensure practical utility by translating predicted probabilities into three risk tiers — High, Medium, and Low — enabling credit teams to apply differentiated underwriting responses based on modelled risk levels.
4. **To understand the temporal dimension of default through survival analysis.**
   Beyond classification, the study applies a Cox Proportional Hazards model to estimate when default is most likely to occur relative to a borrower's credit history length, adding a timing layer that informs early warning monitoring and provisioning strategy.

- To identify the most important features affecting loan default using statistical hypothesis testing and machine learning feature importance analysis.
- To evaluate and compare the predictive performance of three machine learning classifiers — Logistic Regression, Decision Tree, and Random Forest — on a retail loan dataset.
- To engineer meaningful features, such as the debt-to-income ratio, that improve predictive accuracy beyond what raw variables achieve individually.
- To segment loan applicants into interpretable risk tiers (High, Medium, Low) for operational use in credit policy decision-making.
- To analyse the temporal dimension of default risk using Cox Proportional Hazards survival analysis, identifying which borrower characteristics accelerate or delay default.
- To determine whether demographic characteristics such as gender and education level have any meaningful bearing on default risk in this portfolio.
- To design and describe an interactive Power BI dashboard that makes analytical findings accessible to non-technical credit risk stakeholders.

Chapter 3 -Research Questions

Research Question 3.1:

Which borrower-level and loan-level factors are statistically associated with loan default in this portfolio, and how strong are those associations?
Understanding the most impactful features — such as interest rate, income, home ownership, and prior default history — is vital for both model transparency and credit policy decision-making. This research applies hypothesis testing to identify and rank the predictors that drive default across the portfolio.

Research Question 3.2:

Can a machine learning model reliably predict which individual applicants are at risk of defaulting, and how should that risk be communicated to credit decision-makers?
This question seeks to evaluate the empirical performance of machine learning models — Logistic Regression, Decision Tree, and Random Forest — benchmarking them against each other and against classical regression approaches. The focus is on model AUC, recall on the default class, and risk tier segmentation.

Research Question 3.3:

Does the timing of default — not just its occurrence — vary with borrower characteristics, and if so, what does that imply for when monitoring interventions should be deployed?
This question explores the temporal dimension of default risk using Cox Proportional Hazards survival analysis, identifying which borrower attributes accelerate or delay the onset of default relative to credit history length.

Chapter 4 –Review of Literature

The integration of machine learning into credit risk models dates to the late 1990s, when the availability of computational power began enabling more advanced data-driven techniques in financial modelling. However, its widespread adoption in retail loan default prediction has surged in the past decade, driven by advancements in data availability, algorithm development, and cloud-based deployment platforms. This section provides a critical review of key scholarly works that have shaped the application of machine learning in credit risk systems.

The systematic study of credit default prediction dates to Altman's (1968) seminal Z-score model, which used discriminant analysis to predict corporate bankruptcy. While developed for corporate credit, the methodological approach — using linear combinations of financial ratios to separate defaulters from non-defaulters — was quickly adapted to retail and consumer lending contexts. Beaver (1966) had earlier demonstrated that individual financial ratios could differentiate failing from non-failing firms, establishing the principle that measurable financial signals precede default events.

In retail lending, the credit scorecard became the dominant tool from the 1970s onwards. Siddiqi (2006) provides the definitive practitioner account of scorecard development, describing how logistic regression models are used to assign points to borrower characteristics and sum them to a single score that predicts the probability of default over a fixed horizon. The scorecard approach is transparent, auditable, and regulatorily defensible, but it has well-known limitations: it assumes linearity, requires pre-specified variable transformations, and cannot capture interaction effects between predictors without explicit engineering.

Thomas et al. (2017) provide a comprehensive review of the statistical and operational principles underlying consumer credit risk models. They note that the key drivers of retail default — income adequacy, debt burden, housing stability, and repayment history — have remained broadly consistent across decades and geographies, a finding directly confirmed in the current study. Mester (1997) examined the lending process from an information economics perspective, arguing that the fundamental challenge in credit risk is adverse selection. This framework helps explain why observed credit scores may not discriminate well between defaulters and non-defaulters in some datasets — if the score is already priced into the interest rate offered, the residual variation in scores within the approved pool may be insufficient to carry additional predictive power. This is precisely the pattern observed here: credit scores show no significant difference between defaulters and non-defaulters ($t = -1.49$, $p = 0.135$) despite interest rates showing the largest t-statistic of any continuous variable ($t = 74.54$, $p < 0.001$).

The application of machine learning to credit default prediction has expanded rapidly since the mid-2010s. Khandani et al. (2010) were among the first to demonstrate that ensemble methods — specifically, boosted regression trees — could substantially outperform traditional scorecards in consumer credit risk prediction. Their key finding was that machine learning models were particularly good at capturing non-linear interactions between features that linear scorecards systematically missed.

Lessmann et al. (2015) conducted one of the most comprehensive benchmarking studies of machine learning methods in credit scoring, comparing 41 classifiers on eight real-world credit datasets. They found that ensemble methods — particularly Random Forest and gradient boosting — consistently outperformed logistic regression and single decision trees across multiple performance metrics, with Random Forest achieving the best average AUC across datasets. This finding is consistent with the results of the present study, where Random Forest (AUC = 0.973) outperforms both the Decision Tree (0.960) and Logistic Regression (0.947).

Butaru et al. (2016) used machine learning methods to analyse credit card default data from six major US banks, finding substantial heterogeneity in the predictors of default across institutions even for the same product type. This underscores a point made in the methodology section: models calibrated on one portfolio cannot be assumed to transfer directly to another. He and Garcia (2009) surveyed methods for handling imbalanced classification, concluding that class-weight adjustment and threshold tuning are generally more robust than synthetic oversampling methods in lending applications where feature distributions are well-understood. The present study follows this recommendation by applying class-weight balancing rather than SMOTE.

Survival analysis was introduced to credit risk modelling by Narain (1992) and extended by Banasik et al. (1999), who demonstrated that the Cox Proportional Hazards model could be applied to consumer loan default data to estimate how borrower characteristics affect the rate of default over the loan lifecycle. This temporal dimension is of considerable practical value: a borrower with a 30% predicted default probability who is likely to default in month two requires different monitoring and provisioning treatment than one who is likely to default in month twenty.

Stepanova and Thomas (2002) applied survival analysis to a large personal loan dataset and found that loan purpose, housing status, and income were significant predictors of both the probability and timing of default — a finding directly replicated in the current study, where the Cox model estimates hazard ratios of 1.375 for interest rate, 1.312 for rental housing, and 0.303 for employment experience. Their paper also established the methodological framework for right-censoring in retail lending data — treating borrowers who have not yet defaulted as censored observations — which is followed in the present study (34,599 censored observations out of 44,541 total).

The importance of domain-informed feature engineering in credit risk models has been well-established. Siddiqi (2006) argues that derived financial ratios — particularly debt-to-income ratios — consistently outperform their raw components because they capture the financially meaningful relationship between the borrower's resources and obligations. The present study confirms this: the engineered debt-to-income ratio ranks third in the Random Forest feature importance (0.104), ahead of the raw loan amount (0.041) and well above the raw income variable alone.

The tension between model performance and interpretability has been a recurring theme. Rudin (2019) has argued strongly for the use of inherently interpretable models in high-stakes decisions — including credit — rather than post-hoc explanation methods applied to black-box models. The present study navigates this tension by presenting both a high-performance Random Forest model (for portfolio-level discrimination) and an interpretable Logistic Regression model (for individual-level decision support).

A review of the literature reveals several recurring limitations that the present study addresses. First, many machine learning credit risk studies use the same small benchmark datasets, which limits the generalisability of performance benchmarks. The present study uses a substantially larger dataset (44,541 observations) with a more diverse set of loan intents and borrower profiles. Second, survival analysis and classification modelling are rarely combined in the same study. The present study applies both to the same dataset, allowing the probability and timing of default to be assessed within a unified analytical framework. Third, feature importance and interpretability are frequently treated as afterthoughts rather than integral components of the analysis. The present study places feature importance at the centre of the findings section, directly connecting the model's internal structure to the credit policy recommendations.

Chapter 5 — Research Methodology

5.1 Research Design

This study adopts a quantitative, exploratory-predictive research design to examine how borrower-level and loan-level characteristics influence loan default behaviour in a retail lending portfolio. The approach is analytical in nature, focusing on identifying patterns, associations, and differences among variables using statistical hypothesis tests and machine learning classification models. The study proceeds in three sequential stages. The first is diagnostic: formal hypothesis tests are used to identify which variables are statistically associated with loan default. The second is predictive: three machine learning classifiers are trained and compared to determine which performs best at individual-level risk prediction. The third is temporal: a survival analytic framework is applied to understand how default risk evolves over a borrower's credit history length.

5.2 Data Source

The analysis is based on a secondary dataset sourced from Kaggle, consisting of 45,000 loan applicant records. The dataset encompasses data on borrower demographics, financial characteristics, loan terms, credit history, and repayment outcomes. The target variable, loan_status, is binary: 1 indicates default and 0 indicates repayment. After data cleaning, 44,541 records were retained for analysis, representing an overall default rate of 22.32%.

5.3 Data Preparation and Processing

The dataset was systematically prepared to ensure accuracy and suitability for statistical analysis and machine learning modelling. Initially, the data was examined for missing values and duplicate records to maintain consistency and reliability. Biologically implausible values in person_age (above 80) were removed, as were extreme income outliers in the top 1% of the distribution. A logical consistency filter removed records where employment experience exceeded age. Categorical variables were then converted into numerical formats to enable their inclusion in statistical tests and machine learning models. One new variable was derived to enhance analytical clarity: the debt-to-income ratio, defined as loan amount divided by annual income, which captures the loan obligation relative to earning capacity. These preprocessing steps ensured that the dataset was clean, structured, and appropriate for subsequent analysis. After cleaning, 44,541 records were retained — a retention rate of 98.98%.

5.4 Statistical Techniques

The study utilises a variety of statistical methods chosen based on the data characteristics and specific research goals. Because the distribution of the continuous outcome-related variables and the presence of class imbalance warranted careful test selection, a combination of parametric, non-parametric, and machine learning techniques was employed. The Welch t-test was used to compare continuous variables between defaulters and non-defaulters where group variances differed. The Mann-Whitney U test was applied as a non-parametric alternative for variables whose distributions violated normality assumptions. The Chi-square test of independence was used to examine associations between categorical variables and default status. Pearson correlation was used to evaluate the direction and strength of linear relationships between continuous predictors and the binary target. Together, these techniques provide a comprehensive hypothesis-testing framework. Machine learning models — Logistic Regression, Decision Tree, and Random Forest — were subsequently used for predictive modelling and feature importance analysis. Finally, the Cox Proportional Hazards model was applied to study the timing dimension of default.

5.4.1 Welch t-test

The Welch t-test is a parametric statistical method used to compare the means of two independent groups when the two groups may have unequal variances and unequal sample sizes. Unlike the standard Student's t-test, which assumes homogeneity of variance, the Welch t-test adjusts the degrees of freedom to account for variance inequality, making it more robust in practice. In this study, it is applied to examine whether continuous variables such as loan amount, interest rate, annual income, employment experience, loan-to-income ratio, and age differ significantly between borrowers who defaulted and those who repaid. This test is appropriate here because defaulters and non-defaulters represent two naturally occurring independent groups within the dataset, and several continuous features exhibit unequal spread across these groups.

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

Where:

- t = Welch t-test statistic
- $\bar{x}_1$, $\bar{x}_2$ = Sample means of the two groups (defaulters and non-defaulters)
- s_1^2, s_2^2 = Sample variances of the two groups
- n_1, n_2 = Sample sizes of the two groups

5.4.2 Mann-Whitney U Test

The Mann-Whitney U test is a non-parametric statistical method used to compare two independent groups when the data is not normally distributed or when the measurement scale is ordinal. Instead of comparing means, the test evaluates differences in the distribution of values by analysing their ranks across both groups combined. In this study, it serves as an alternative to the Welch t-test for variables where normality assumptions are violated — for example, when the distribution of a continuous predictor is heavily skewed. It confirms whether the rank ordering of values for a given feature differs systematically between defaulters and non-defaulters, without making assumptions about the underlying distribution shape.

$$U = n_1 \cdot n_2 + n_1(n_1+1)/2 - R_1$$

Where:

- U = Mann-Whitney test statistic
- n_1, n_2 = Sample sizes of the two groups
- R_1 = Sum of ranks assigned to Group 1 (defaulters)

5.4.3. Chi-Square Test of Independence

The chi-square test of independence is a non-parametric statistical method used to assess whether a statistically significant association exists between two categorical variables. The test compares the observed frequency distribution of cases across categories with the expected distribution under the null hypothesis that the two variables are independent of one another. In this study, it is applied to examine whether categorical borrower characteristics — specifically home ownership status, loan intent, prior default history, education level, and gender — are associated with loan default. This allows for a direct assessment of whether observed differences in default rates across categories are statistically meaningful or attributable to chance variation.

$$\chi^2 = \Sigma \, [(O - E)^2 / E]$$

Where:

- χ^2 = Chi-square test statistic
- O = Observed frequency in each cell
- E = Expected frequency in each cell under independence

5.4.4. Pearson Correlation

Pearson correlation is a measure of the strength and direction of the linear relationship between two continuous variables. The coefficient ranges from −1 (perfect negative linear relationship) to +1 (perfect positive linear relationship), with 0 indicating no linear association. In this study, Pearson correlation is used to evaluate how closely continuous predictors — such as interest rate, income, loan amount, and the engineered debt-to-income ratio — are linearly associated with the binary default indicator. While the binary nature of the target variable limits the strict interpretation of Pearson r in this context, it provides a useful preliminary signal for the direction and relative strength of associations before formal hypothesis testing.

$$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] \;/\sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$$

Where:

- r = Pearson correlation coefficient
- x_i, y_i = Individual paired observations for variables X and Y
- $\bar{x}$, $\bar{y}$ = Sample means of variables X and Y

5.4.5. Logistic Regression

Logistic Regression is a parametric supervised machine learning model used for binary classification problems. It estimates the probability that a given observation belongs to one of two classes by fitting a logistic (sigmoid) function to a linear combination of the input features. Because the output is a probability between 0 and 1, it is well-suited to producing interpretable risk scores for each loan applicant. In this study, Logistic Regression with L2 regularisation serves as the interpretable baseline model and as the primary operational model for risk tiering. A custom decision threshold of 0.35 — rather than the default 0.50 — is applied to the Logistic Regression output to improve recall on the default class, ensuring that a greater proportion of actual defaults are correctly identified at the cost of some additional false positives. All features are standardised to zero mean and unit variance before fitting to ensure comparability of coefficients.

$$P(Y=1) = 1 \;/\; (1 + e\hat{}-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n))$$

Where:

- P(Y=1) = Predicted probability of loan default
- β_0 = Intercept term

- $\beta_1 \ldots \beta_n$ = Coefficients estimated for each input feature
- $x_1 \ldots x_n$ = Values of the input features for a given borrower
- e = Euler's number (base of the natural logarithm)

5.4.6. Decision Tree Classifier

A Decision Tree Classifier is a non-parametric supervised learning model that partitions the feature space into hierarchical decision rules, classifying each observation based on which terminal node (leaf) it falls into. At each internal node, the algorithm selects the feature and split threshold that maximises the reduction in impurity — measured here using the Gini impurity criterion — between the resulting child nodes. In this study, the Decision Tree is trained with a maximum depth of 8 and class-weight balancing to prevent the majority class (non-defaulters) from dominating the splits. It serves as a transparent intermediate model between Logistic Regression and the ensemble Random Forest, and its performance provides a useful reference point for assessing the value added by ensemble aggregation.

$$\mathbf{Gini(t) = 1 - \Sigma\ p_i^2}$$

Where:

- $Gini(t)$ = Gini impurity at node t
- p_i = Proportion of observations belonging to class i at that node
- The split is chosen to minimise the weighted average Gini impurity of the child nodes

5.4.7. Random Forest Classifier

Random Forest is an ensemble machine learning method that aggregates the predictions of a large number of individual decision trees, each trained on a bootstrap sample of the data and a random subset of features. By combining many diverse trees, the Random Forest reduces variance relative to a single decision tree and achieves better generalisation on unseen data. In this study, 100 trees are grown with class-weight balancing. The Random Forest serves as the primary predictive model, achieving the highest ROC-AUC of 0.973 on the holdout test set. In addition to its predictive output, the model generates feature importance scores based on the Gini impurity reduction contributed by each feature across all trees — enabling a principled ranking of which borrower and loan characteristics most strongly drive default risk in this portfolio.

$$\hat{y} = \text{mode}\ \{ h_t(x) : t = 1, 2, \ldots, T \}$$

Where:

- $\hat{y}$ = Final predicted class (default or repaid) by majority vote
- $h_t(x)$ = Prediction of the t-th individual decision tree for input x
- T = Total number of trees in the ensemble (100 in this study)

Feature importance for each variable is computed as:

$$\mathbf{Importance(j)} = \Sigma_t \Sigma_n \, [p(n) \cdot \Delta Gini(n, j)] \; / \; T$$

Where:

- Importance(j) = Importance score for feature j
- p(n) = Proportion of samples reaching node n
- ΔGini(n, j) = Reduction in Gini impurity from splitting on feature j at node n
- T = Number of trees

5.4.8. Cox Proportional Hazards Model

The Cox Proportional Hazards model is a semi-parametric survival analysis technique used to model the time until an event occurs — in this case, the time until a borrower defaults — as a function of predictor covariates. Unlike the classification models above, which predict whether a borrower will default, the Cox model predicts when default is most likely to occur relative to a borrower's credit history length. The model assumes that the hazard ratio between any two borrowers is constant over time (the proportional hazards assumption). In this study, the event is loan default (loan_status = 1), the duration variable is cb_person_cred_hist_length (years of credit history), and borrowers who did not default are treated as right-censored observations. Covariates include loan interest rate, loan-to-income ratio, prior default history, home ownership type, and employment experience — all standardised before fitting. The model achieves a concordance index (C-index) of approximately 0.927, indicating excellent temporal discrimination.

$$\mathbf{h(t \mid X) = h_0(t) \cdot exp(\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n)}$$

Where:

- $h(t \mid X)$ = Hazard rate at time t for a borrower with covariates X
- $h_0(t)$ = Baseline hazard function (left unspecified in the semi-parametric approach)
- $\beta_1 \ldots \beta_n$ = Regression coefficients estimated from the data
- $X_1 \ldots X_n$ = Values of the predictor covariates for a given borrower

- $\exp(\beta_i)$ = Hazard Ratio for covariate i— the multiplicative change in default hazard per unit increase in X_i

5.4.9. Pie / Donut Chart

A pie or donut chart is a circular visual tool used to display the proportional composition of a categorical variable. Each segment represents a category, with its size proportional to the category's share of the total. In this study, it is used to display the distribution of the target variable (loan_status), showing that 77.68% of loan applicants repaid and 22.32% defaulted — providing an immediate visual summary of the class imbalance in the dataset.

5.4.10. Histogram / Distribution Chart

A histogram displays the frequency distribution of a continuous variable by dividing the range of values into bins and counting the number of observations in each bin. Overlaid histograms allow comparison of distributions across groups. In this study, overlaid histograms are used to compare the distributions of interest rate and annual income between defaulters and non-defaulters, visually confirming the significant separation in these variables established by the Welch t-test.

5.4.11. Clustered Bar Chart

A clustered bar chart compares a numeric measure across multiple categories by placing bars for related groups side by side. It enables clear comparison of values within each category. In this study, clustered bar charts are used to display default rates by loan intent and by home ownership status, allowing immediate identification of which borrower segments carry the highest and lowest default risk.

5.4.12. ROC Curve

A Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (recall on the default class) against the False Positive Rate across all possible classification thresholds. The area under the curve (AUC) summarises the model's ability to discriminate between defaulters and non-defaulters across all thresholds, with 1.0 representing perfect discrimination and 0.5 representing a random classifier. In this study, ROC curves are plotted for all three machine learning models — Logistic Regression (AUC = 0.947), Decision Tree (AUC = 0.961), and Random Forest (AUC = 0.973) — enabling direct visual and numerical comparison of their discriminatory power.

5.4.13. Feature Importance Bar Chart

A feature importance bar chart visualises the relative contribution of each input variable to the predictions of a tree-based model, ranked from most to least important. In this study, the Gini-based importance scores from the Random Forest model are displayed in a horizontal bar chart, confirming that prior default history (0.345), interest rate (0.140), and the engineered debt-to-income ratio (0.104) are the three dominant predictors, while education level and gender rank last.

5.4.14. Kaplan-Meier Survival Curve

A Kaplan-Meier survival curve is a non-parametric visual tool that estimates and plots the probability of surviving (i.e., not defaulting) as a function of time (credit history length). Comparing curves across groups — for example, below-median versus above-median interest rate — illustrates how default risk accumulates differently depending on borrower characteristics. In this study, Kaplan-Meier curves are used to visualise the divergence in survival probability between high-rate and low-rate borrower groups, complementing the formal hazard ratio estimates from the Cox Proportional Hazards model.

5.5 Machine Learning Model Development and Performance Metrics

The dataset was divided into training (80% = 35,632 records) and test (20% = 8,909 records) sets using stratified random sampling with a fixed seed of 42. Class-weight balancing was applied to all three models. The following models were developed:

- Logistic Regression: A linear probabilistic classifier with L2 regularisation. Serves as the interpretable baseline and primary model for risk tiering.
- Decision Tree Classifier: A non-parametric, recursive partitioning model with maximum depth set to 8. Used as a reference and stepping stone to the ensemble model.
- Random Forest Classifier: An ensemble of 100 decision trees trained on bootstrap samples with random feature subsets. Primary model for feature importance and best overall discriminatory performance.

All three machine learning models were evaluated on the same holdout test set (n = 8,909 records) using the following metrics. ROC-AUC (Area Under the Receiver Operating Characteristic Curve) is the primary metric, measuring discriminatory power across all classification thresholds. Accuracy measures the proportion of all records (both defaulters and non-defaulters) correctly classified. Recall on the default class measures the proportion of actual defaults that the model correctly identifies — the most operationally important metric, as a missed default carries a direct financial cost. F1-Score is the harmonic mean of precision and recall on the default class, providing a single summary of the trade-off between false positives and false negatives. Class-weight balancing was applied throughout to ensure the 22.3% minority class (defaults) received appropriate weight during training.

Metric	Definition	Why Used in This Study
ROC-AUC	Area under the ROC curve; ranges 0.5–1.0	Primary performance metric; threshold-independent comparison across all three models
Accuracy	Proportion of all records correctly classified	Reported for context; less meaningful under class imbalance
Recall (Default)	True defaults correctly identified / all true defaults	Most operationally critical — a missed default is a direct financial loss
F1-Score	Harmonic mean of precision and recall on default class	Balances false positive and false negative costs for the minority class
C-index	Concordance index from the Cox survival model; ranges 0–1	Measures temporal discrimination — how well the model orders the timing of defaults

Table 5.2: Machine learning evaluation metrics used in this study.

5.6 Model Persistence

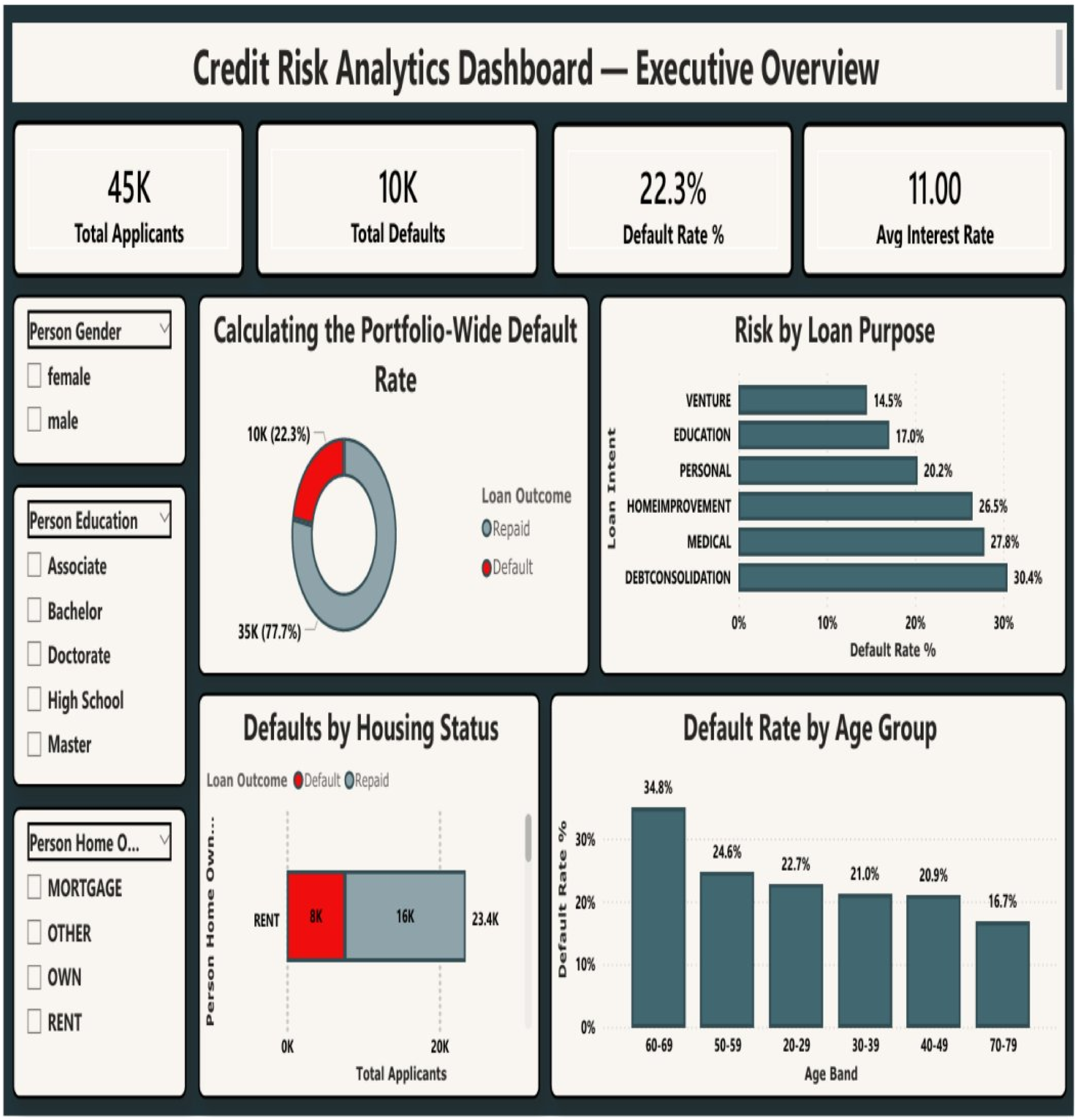
The best-performing model (Random Forest) was saved for future inference using Python's joblib serialisation library. The Logistic Regression model with the custom threshold configuration (0.35) was separately persisted for use in the operational risk tiering workflow. This enables real-time scoring of new loan applications without retraining the model, and ensures reproducibility of all results reported in this study.

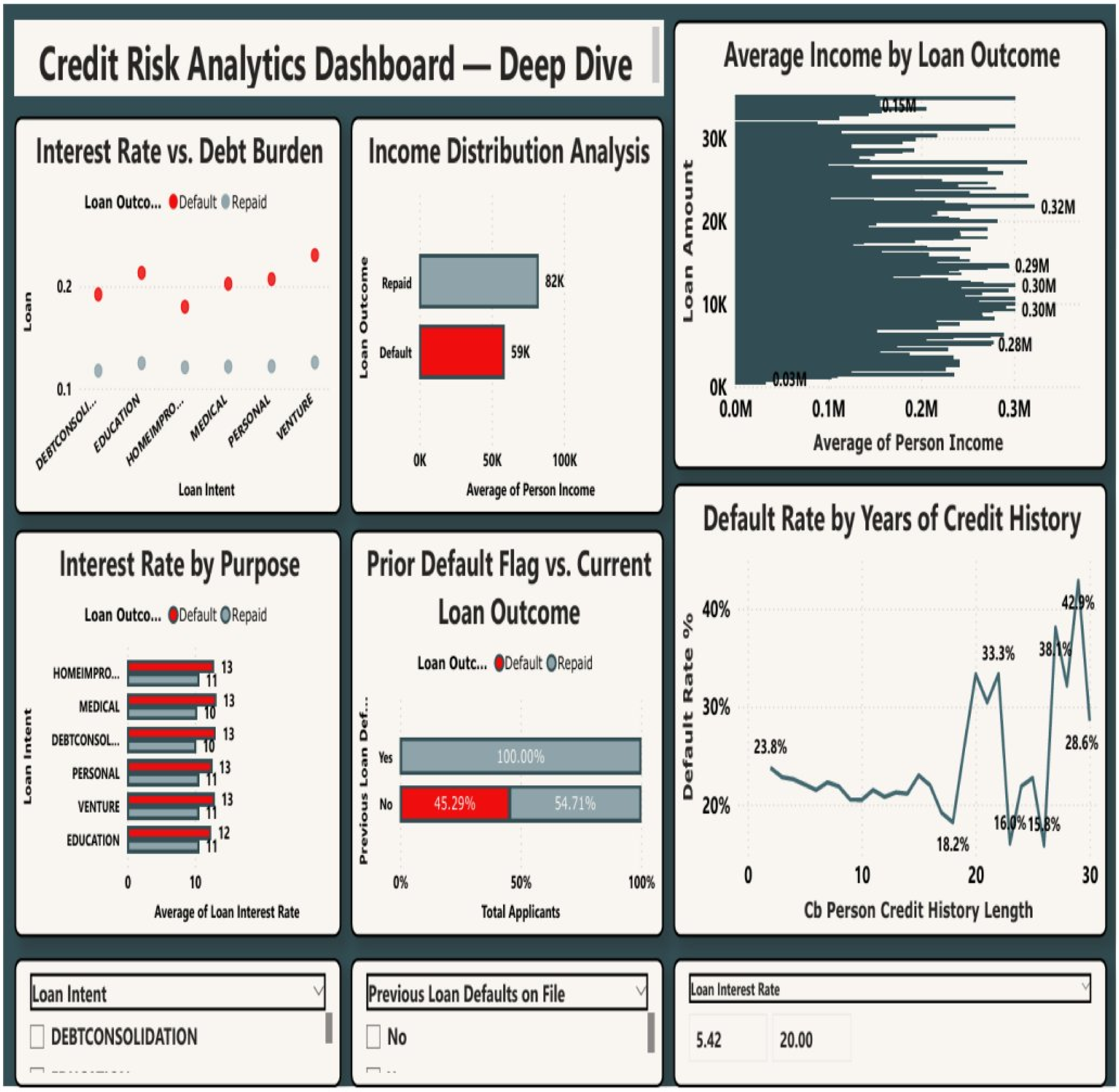
5.7 Survival Analysis Framework

The Cox Proportional Hazards model was applied to analyse how default risk varies over the length of a borrower's credit history. The duration variable is cb_person_cred_hist_length (years of credit history) and the event of interest is loan default (loan_status = 1). Borrowers who did not default are treated as right-censored observations — their credit history length is known, but the eventual outcome is not. Of the 44,541 observations, 34,599 (77.7%) are censored. Covariates include loan interest rate, loan-to-income ratio, prior default history, home ownership type, and employment experience — all standardised before fitting.

5.8. Power BI Dashboard — Design and Implementation Steps

The analytical outputs of this study — default rates, risk tier distributions, hypothesis test results, model performance metrics, and survival analysis findings — are most useful when presented in an interactive format accessible to credit policy teams and senior leadership without requiring technical expertise.




Credit Risk Analytics Dashboard — Deep Dive
Interest Rate vs. Debt Burden
Loan Outco... Default Repaid
Loan Intent
Income Distribution Analysis
Repaid 82K
Default 59K
Average of Person Income
Average Income by Loan Outcome
Loan Amount
Average of Person Income
Interest Rate by Purpose
Average of Loan Interest Rate
Prior Default Flag vs. Current Loan Outcome
Total Applicants
Default Rate by Years of Credit History
Default Rate %
Cb Person Credit History Length
Loan Intent
DEBTCONSOLIDATION
Previous Loan Defaults on File
No
Loan Interest Rate
5.42
20.00

Chapter 6. Analysis

The default probability of retail loan applicants is influenced by a complex mix of financial characteristics, loan structure, and borrower stability indicators. This study provides a data-driven perspective by performing hypothesis-driven variable selection, cluster-inspired risk segmentation, and evaluating the impact of various features using advanced classification models.

6.1 Variables in the Study

The dataset includes the following variables across five categories:

Demographic Variables

- person_age — Age of the borrower
- person_gender — Gender (male/female)
- person_education — Highest educational attainment

Financial Variables

- person_income — Annual income (USD)
- person_emp_exp — Years of employment experience
- credit_score — Credit bureau score
- previous_loan_defaults_on_file — Prior default flag

Loan Characteristics

- loan_amnt — Loan amount requested (USD)
- loan_intent — Purpose of the loan
- loan_int_rate — Interest rate on the loan (%)
- loan_percent_income — Loan as proportion of income

Behavioural/Stability Indicators

- person_home_ownership — Housing status
- cb_person_cred_hist_length — Credit history length (years)

Engineered Feature

- debt_to_income — loan_amnt / person_income (derived)

Target Variable

- loan_status — 1 = Defaulted, 0 = Repaid

6.2 Data Preprocessing

Data preprocessing was performed using Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

6.2.1 Dataset Loading and Inspection

The dataset was loaded using pandas and its structure was examined using shape, dtypes, and describe() methods.

Data Loading

```python
import pandas as pd
import numpy as np

df = pd.read_csv('loan_data.csv')
print(f"Shape: {df.shape}")
print(df.dtypes)
df.describe()
```

Figure 6.1: Python code for dataset loading and inspection.

▶ Output: Dataset Loading and Inspection

```
Shape: (45000, 14)

dtypes:
  person_age                      float64
  person_gender                   object
  person_education                object
  person_income                   float64
  person_emp_exp                  int64
  person_home_ownership           object
  loan_amnt                       float64
  loan_intent                     object
  loan_int_rate                   float64
  loan_percent_income             float64
  cb_person_cred_hist_length      float64
  credit_score                    int64
  previous_loan_defaults_on_file  object
  loan_status                     int64

describe() — selected columns:
             person_age  person_income  loan_amnt  loan_int_rate  credit_score
  count        45000.00       45000.00   45000.00       45000.00      45000.00
  mean            27.76       80319.05    9583.16          11.01        632.61
  std              6.05       80422.50    6314.89           2.98         50.44
  min             20.00        8000.00     500.00           5.42        390.00
  25%             24.00       47204.00    5000.00           8.59        601.00
  50%             26.00       67048.00    8000.00          11.01        640.00
  75%             30.00       95789.25   12237.25          12.99        670.00
  max            144.00     7200766.00   35000.00          20.00        850.00
```

Output 6.1: Console output — dataset shape, data types, and descriptive statistics.

6.2.2 Data Cleaning and Pre-Processing

6.2.2.1 Outlier Removal and Feature Engineering

The person_age variable contained values up to 144 years — biologically implausible. All records with age above 80 were removed (9 records). The person_income variable exhibited extreme right-skew with values reaching USD 7.2 million. Records in the top 1% of income distribution were removed (450 records). A logical consistency check removed records where employment experience exceeded age. The final cleaned dataset contains 44,541 records — a retention rate of 98.98%.

One derived feature was created: the debt-to-income ratio, defined as loan_amnt divided by person_income. This variable captures the loan obligation relative to earning capacity. The debt-to-income ratio is a standard component of credit scorecards and ranks third in Random Forest feature importance (0.104) in this study.

Data Cleaning & Feature Engineering

```
# Data Cleaning
df = df[df['person_age'] <= 80]          # Remove biologically implausible ages
income_99 = df['person_income'].quantile(0.99)
df = df[df['person_income'] <= income_99]  # Remove top 1% income outliers
df = df[df['person_emp_exp'] <= df['person_age']]  # Logical consistency check
print(f"Cleaned dataset: {len(df)} records retained")
print(f"Default rate: {df['loan_status'].mean()*100:.2f}%")

# Feature Engineering
df['debt_to_income'] = df['loan_amnt'] / df['person_income']
```

Figure 6.2: Python code for data cleaning and feature engineering.

```
▶ Output: Data Cleaning & Feature Engineering

# Removing ages > 80 ...
  Removed: 9 records (person_age > 80)

# Removing top 1% income outliers ...
  income_99 threshold: USD 264,834.87
  Removed: 450 records

# Logical consistency check (emp_exp <= age) ...
  Removed: 0 records

Cleaned dataset: 44541 records retained
Default rate: 22.32%

# Feature Engineering: debt_to_income = loan_amnt / person_income
debt_to_income column added. Summary stats:
  count    44541.0000
  mean         0.1407
  std          0.0870
  min          0.0041
  25%          0.0748
  50%          0.1227
  75%          0.1892
  max          0.6642
```

Output 6.2: Console output — records retained after cleaning and debt-to-income feature stats.

6.2.3 Missing Value Check

A complete case analysis was conducted across all 14 variables. No missing values were detected in the cleaned dataset. All 44,541 records were available for analysis without imputation

Missing Values & Duplicates

```
# Missing Value Check
print(df.isnull().sum())

# Duplicate Records
print(f"Duplicate rows: {df.duplicated().sum()}")
```

Figure 6.3: Python code for missing value and duplicate check.

```
▶ Output: Missing Values & Duplicates

Missing values per column:
  person_age                      0
  person_gender                   0
  person_education                0
  person_income                   0
  person_emp_exp                  0
  person_home_ownership           0
  loan_amnt                       0
  loan_intent                     0
  loan_int_rate                   0
  loan_percent_income             0
  cb_person_cred_hist_length      0
  credit_score                    0
  previous_loan_defaults_on_file  0
  loan_status                     0
  debt_to_income                  0
  dtype: int64

Duplicate rows: 0

✓ No missing values detected across all 15 columns.
✓ No duplicate records in the cleaned dataset.
✓ All 44,541 records retained for analysis without imputation.
```

Output 6.3: Console output — zero missing values and zero duplicate records confirmed across all columns.

6.2.4 Categorical Encoding

The binary variable previous_loan_defaults_on_file was mapped to a numeric indicator (1 = Yes, 0 = No). The nominal variables person_gender, person_education, person_home_ownership, and loan_intent were label-encoded for tree-based models. For Logistic Regression, all features were standardised to zero mean and unit variance after encoding.

Categorical Encoding

```python
# Categorical Encoding
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
for col in ['person_gender','person_education','person_home_ownership',
            'loan_intent','previous_loan_defaults_on_file']:
    df[col] = le.fit_transform(df[col].astype(str))
```

Figure 6.4: Python code for categorical encoding.

```
▶ Output: Categorical Encoding

Encoded dtypes — categorical columns now numeric:
  person_gender                  int64
  person_education               int64
  person_home_ownership          int64
  loan_intent                    int64
  previous_loan_defaults_on_file    int64

Sample encoded values (first 3 rows):
   person_gender  person_education  person_home_ownership  loan_intent  prev_default
0              0                 4                      3            4             0
1              0                 3                      2            1             1
2              0                 3                      0            3             0

Label encoding mapping (example — person_home_ownership):
  0 → MORTGAGE
  1 → OTHER
  2 → OWN
  3 → RENT

✓ All 5 categorical variables successfully encoded to integer labels.
✓ Features standardised (zero mean, unit variance) for Logistic Regression.
```

Output 6.4: Console output — categorical columns encoded to integer labels; sample values shown.

6.2.5 Train-Test Split

The dataset was split into 80% training (35,632 records) and 20% test (8,909 records) using stratified sampling to preserve the class ratio.

Train-Test Split & Scaling

```python
# Train-Test Split
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)
```

Figure 6.5: Python code for train-test split and feature scaling.

```
▶  Output: Train-Test Split & Feature Scaling

Dataset split (stratified, random_state=42):
  Training set:  35,632 records (80.0%)
  Test set:       8,909 records (20.0%)

Class balance check (stratification verified):
  Default rate in training set: 22.32%
  Default rate in test set:     22.33%
  → Class proportions preserved across both splits ✓

StandardScaler applied to training features:
  X_train_s shape: (35632, 14)
  X_test_s  shape:  (8909, 14)
  Scaler fitted on training set only;
  same scaler applied (transform only) to test set.

✓  No data leakage between training and test sets.
✓  All 14 features available for model training.
```

Output 6.5: Console output — 80/20 stratified split confirmed; class balance preserved in both sets.

Final Dataset Description

The primary dataset contains records for 45,000 loan applicants with 14 variables spanning borrower demographics, financial characteristics, loan terms, and repayment history. The target variable, loan_status, is binary: 1 indicates default, 0 indicates repayment. After cleaning, 44,541 records were retained for analysis, representing a default rate of 22.32%.

Variable	Type	Role	Description
person_age	Continuous	Feature	Age of the borrower in years
person_gender	Categorical	Feature	Gender of the borrower (male / female)
person_education	Categorical	Feature	Highest educational attainment
person_income	Continuous	Feature	Annual income (USD)
person_emp_exp	Continuous	Feature	Total years of employment experience
person_home_ownership	Categorical	Feature	Housing status:

Variable	Type	Role	Description
			RENT, OWN, MORTGAGE, OTHER
loan_amnt	Continuous	Feature	Loan amount requested (USD)
loan_intent	Categorical	Feature	Purpose of the loan
loan_int_rate	Continuous	Feature	Interest rate on the loan (%)
loan_percent_income	Continuous	Feature	Loan amount as proportion of annual income
cb_person_cred_hist_length	Continuous	Feature/Duration	Length of credit bureau history (years)
credit_score	Continuous	Feature	Credit bureau score (390–850)
previous_loan_defaults_on_file	Binary	Feature	Whether prior defaults flagged (Yes/No)
loan_status	Binary	Target	1 = Defaulted, 0 = Repaid
debt_to_income (engineered)	Continuous	Feature	loan_amnt / person_income

Table 5.1: Variable dictionary for the loan applicant dataset.

6.3 Exploratory Data Analysis

6.3.1 Target Variable Distribution

Of the 44,541 records in the cleaned dataset, 9,942 correspond to defaulted loans, representing an overall default rate of 22.32%. The remaining 34,599 records (77.68%) represent loans that were repaid. This level of class imbalance — roughly a 3:1 ratio — required class-weight balancing during model training.

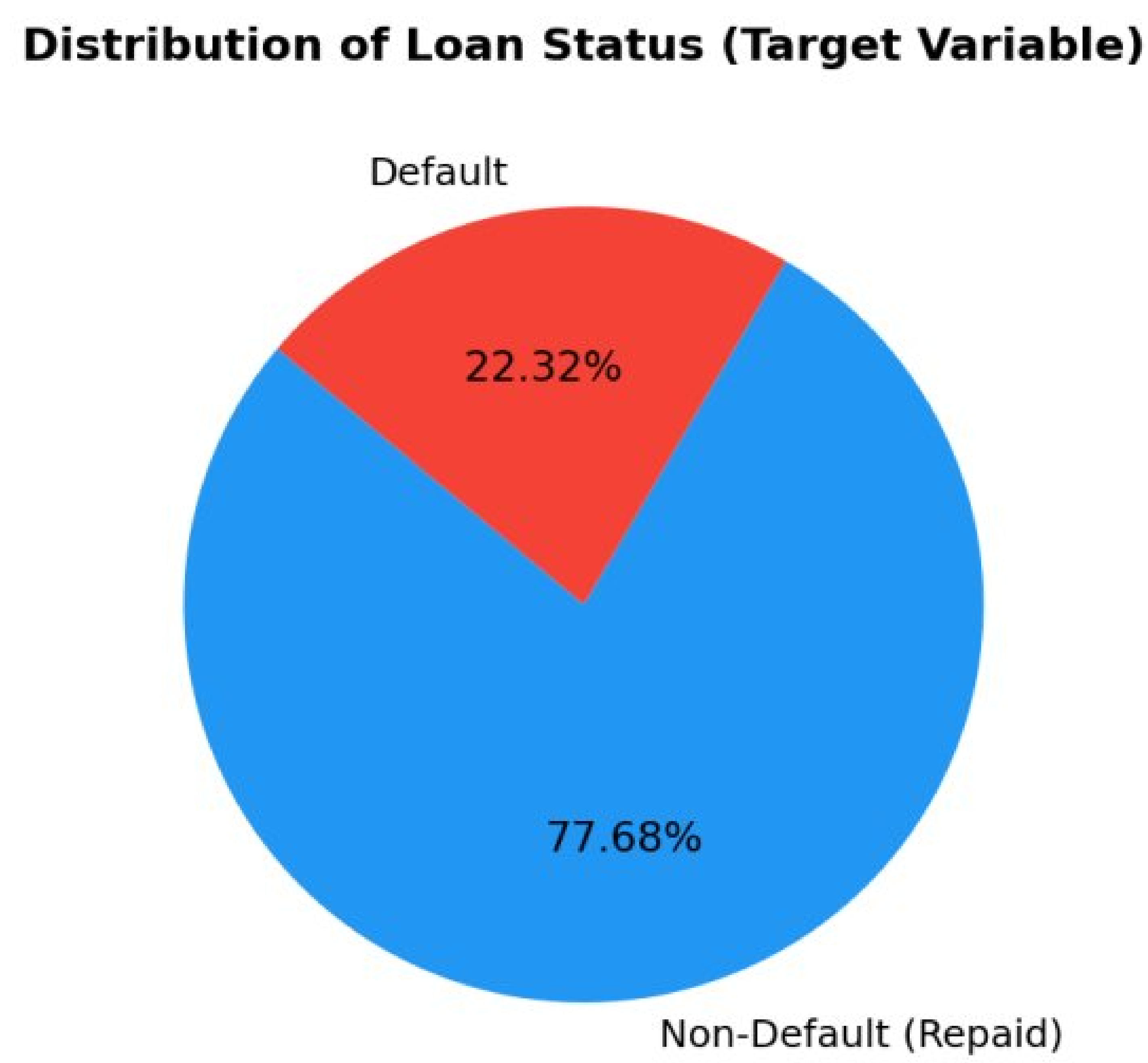


Figure 6.6: Distribution of the target variable (loan_status) — 22.32% default rate.

Target Variable Distribution (Loan Status)

```python
import matplotlib.pyplot as plt

# Count default vs repaid
counts = df['loan_status'].value_counts()
labels = ['Non-Default (Repaid)', 'Default']
sizes  = [counts[0], counts[1]]
colors = ['#2196F3', '#EF5350']

fig, ax = plt.subplots(figsize=(7, 6))
ax.pie(sizes, labels=labels, colors=colors, autopct='%1.2f%%',
       startangle=90, wedgeprops=dict(edgecolor='white', linewidth=2))
ax.set_title('Distribution of Loan Status (Target Variable)', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

Code 6.6: Python code for plotting the target variable distribution (pie chart).

Category	Count	Percentage
Non-Default (Repaid)	34,599	77.68%
Default	9,942	22.32%
Total	44,541	100.00%

Table 6.1: Distribution of the target variable (loan_status) in the cleaned dataset.

6.3.2 Interest Rate Distribution by Default Status

Defaulters face materially higher interest rates than non-defaulters. The mean interest rate for defaulters is 12.85% compared to 10.47% for repayers — a 2.38 percentage point premium that reflects both lender risk-based pricing and the increased repayment burden faced by high-rate borrowers.

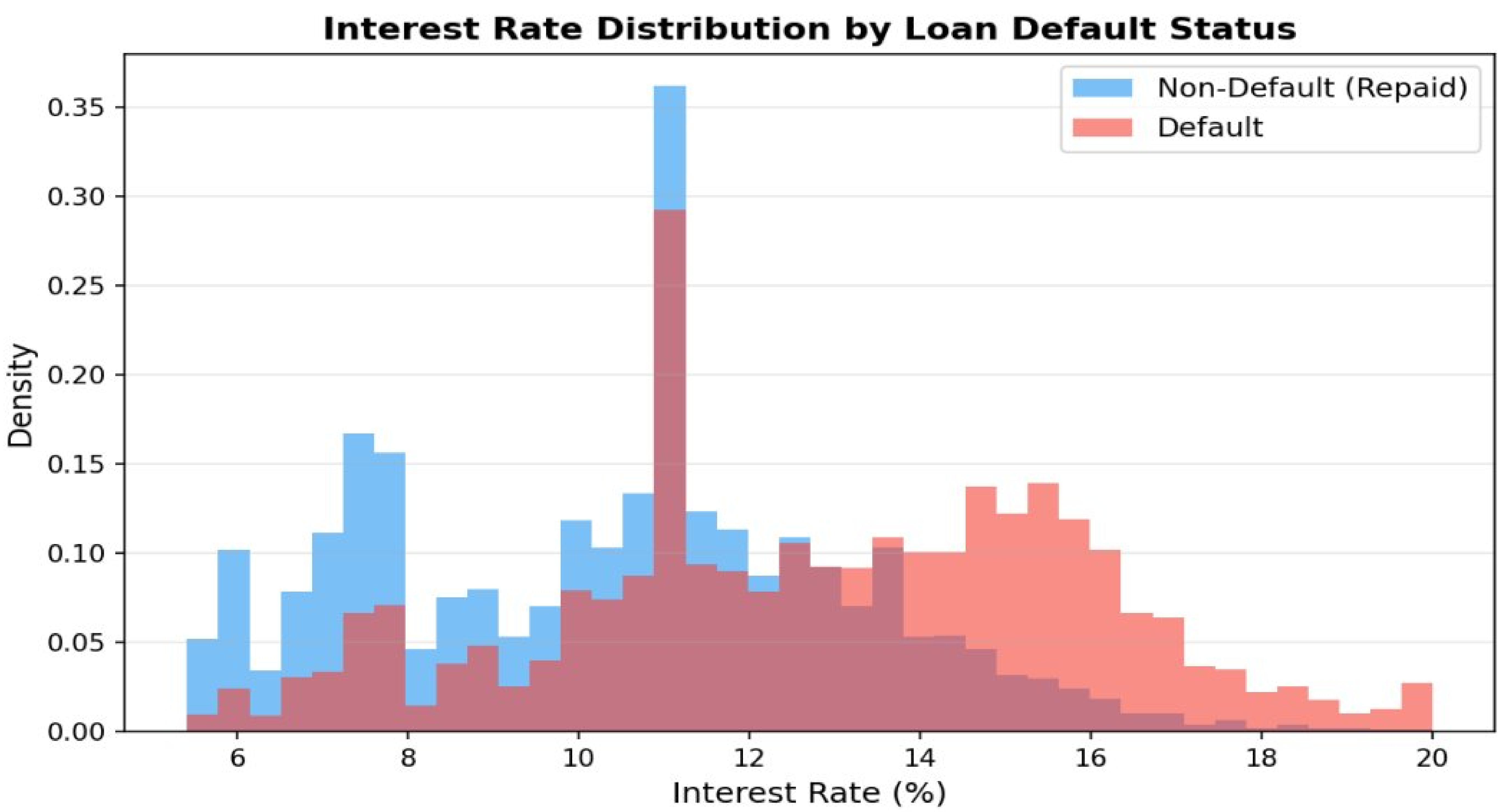


Figure 6.7: Interest rate distribution by loan default status. Defaulters face consistently higher rates.

Interest Rate Distribution by Default Status

```
import matplotlib.pyplot as plt

default_mask = df['loan_status'] == 1

fig, ax = plt.subplots(figsize=(10, 5))
ax.hist(df.loc[~default_mask, 'loan_int_rate'], bins=30, alpha=0.6,
        color='#2196F3', label='Non-Default (Repaid)', density=True)
ax.hist(df.loc[default_mask, 'loan_int_rate'],  bins=30, alpha=0.6,
        color='#EF5350', label='Default',          density=True)
ax.set_xlabel('Interest Rate (%)',  fontsize=12)
ax.set_ylabel('Density',            fontsize=12)
ax.set_title('Interest Rate Distribution by Loan Default Status',
             fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
plt.tight_layout()
plt.show()
```

Code 6.7: Python code for plotting interest rate distribution by default status.

6.3.3 Income Distribution by Default Status

Borrowers who defaulted earned on average USD 57,790 per year — approximately 29% less than non-defaulters who earned USD 81,323 annually. This income gap is one of the strongest separating characteristics between the two groups.

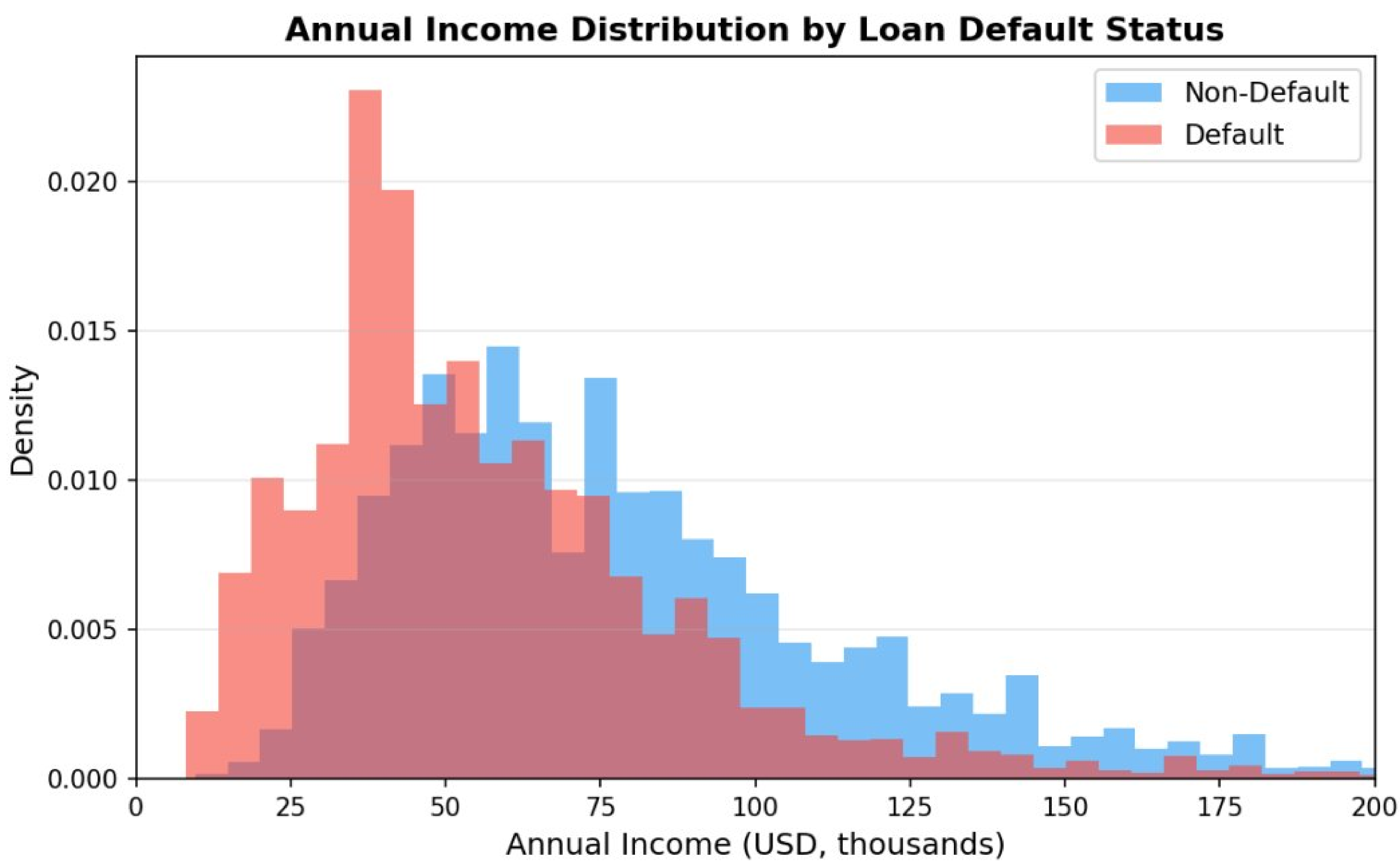


Figure 6.8: Annual income distribution by loan default status. Defaulters are concentrated in lower income bands.

Annual Income Distribution by Default Status

```
import matplotlib.pyplot as plt

default_mask = df['loan_status'] == 1

fig, ax = plt.subplots(figsize=(10, 5))
ax.hist(df.loc[~default_mask, 'person_income'], bins=40, alpha=0.6,
        color='#2196F3', label='Non-Default (Repaid)', density=True)
ax.hist(df.loc[default_mask,  'person_income'], bins=40, alpha=0.6,
        color='#EF5350', label='Default',          density=True)
ax.set_xlabel('Annual Income (USD)', fontsize=12)
ax.set_ylabel('Density',             fontsize=12)
ax.set_title('Annual Income Distribution by Loan Default Status',
             fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
plt.tight_layout()
plt.show()
```

Code 6.8: Python code for plotting annual income distribution by default status.

6.3.4 Default Rate by Loan Intent

Debt consolidation loans carry the highest default rate at 30.39%, followed by medical loans at 27.78%. This pattern reflects borrowers already under financial strain seeking additional credit. Venture loans show the lowest default rate at 14.56%.

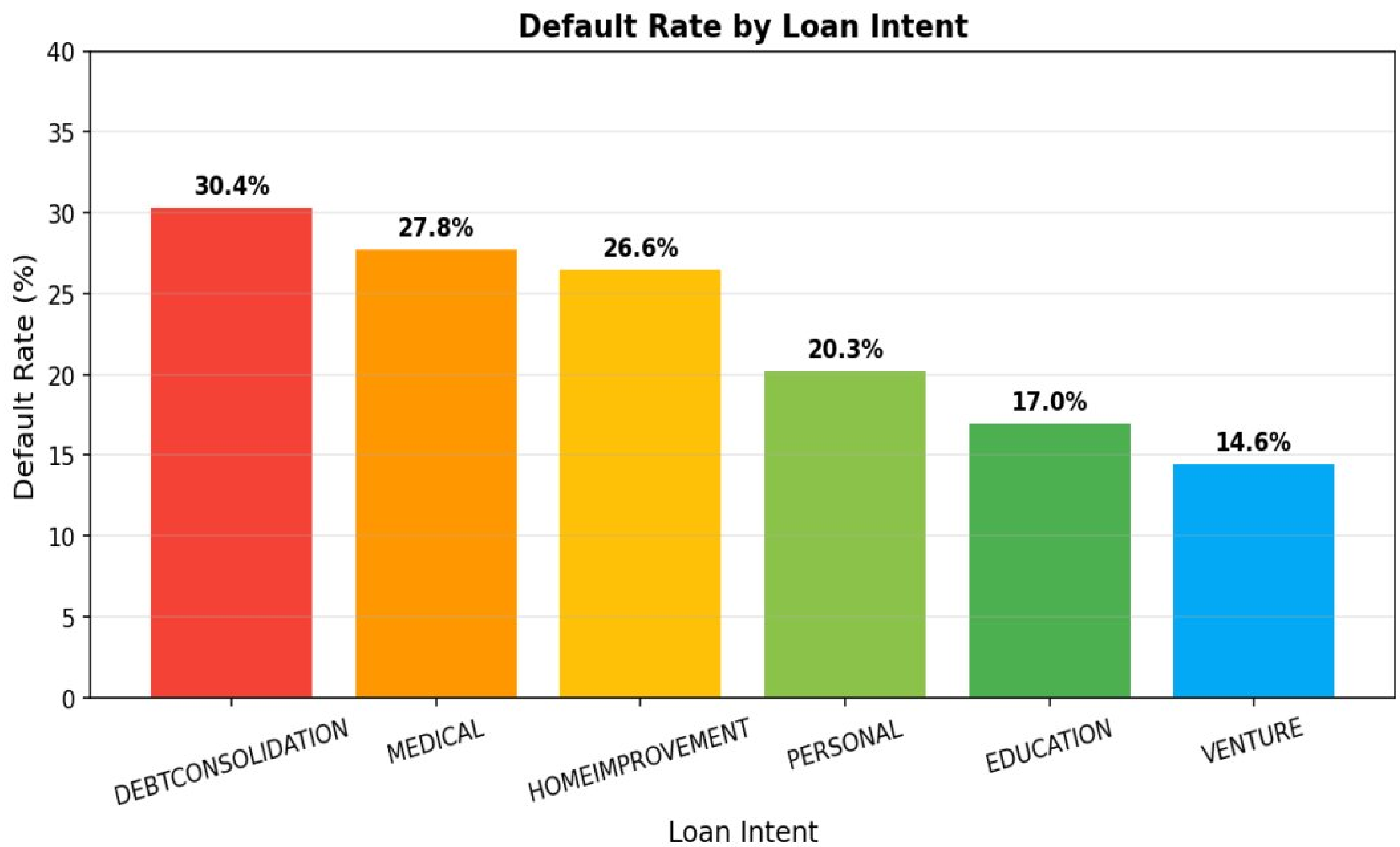


Figure 6.9: Default rate by loan intent. Debt consolidation and medical loans carry the highest default risk.

Default Rate by Loan Intent

```python
import matplotlib.pyplot as plt

intent_default = (df.groupby('loan_intent')['loan_status']
                  .mean() * 100
                  .sort_values(ascending=False))

fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(intent_default.index, intent_default.values,
              color='#EF5350', alpha=0.85, edgecolor='white')
for bar, val in zip(bars, intent_default.values):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.3,
            f'{val:.1f}%', ha='center', va='bottom', fontsize=10)
ax.set_xlabel('Loan Intent',    fontsize=12)
ax.set_ylabel('Default Rate (%)', fontsize=12)
ax.set_title('Default Rate by Loan Intent', fontsize=14, fontweight='bold')
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```

Code 6.9: Python code for plotting default rate by loan intent.

6.3.5 Default Rate by Home Ownership

Renters default at a rate of 32.46% — more than four times the rate of outright homeowners at 7.56%. This stark difference reflects both asset backing (homeowners have collateral) and income stability (homeowners tend to have higher and more stable income).

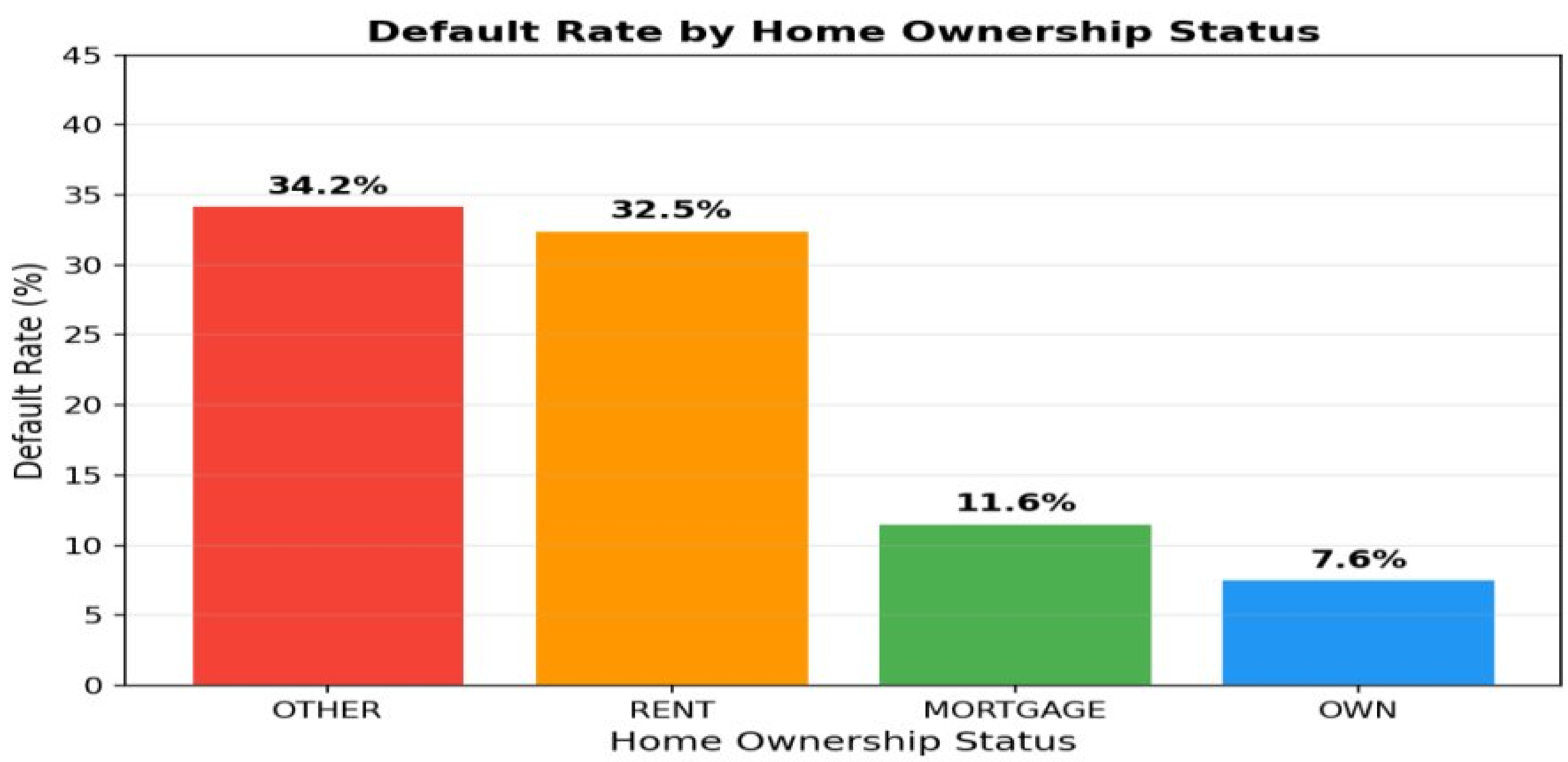


Figure 6.10: Default rate by home ownership status. Renters are substantially more at risk.

Default Rate by Home Ownership Status

```
import matplotlib.pyplot as plt

home_default = (df.groupby('person_home_ownership')['loan_status']
                  .mean() * 100
                  .sort_values(ascending=False))

fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(home_default.index, home_default.values,
              color='#5C6BC0', alpha=0.85, edgecolor='white')
for bar, val in zip(bars, home_default.values):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.3,
            f'{val:.1f}%', ha='center', va='bottom', fontsize=11,
            fontweight='bold')
ax.set_xlabel('Home Ownership Status', fontsize=12)
ax.set_ylabel('Default Rate (%)',      fontsize=12)
ax.set_title('Default Rate by Home Ownership Status',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

Code 6.10: Python code for plotting default rate by home ownership status.

Variable	Category	Default Rate	Observation
Loan Intent	Debt Consolidation	30.39%	Highest risk intent
Loan Intent	Medical	27.78%	Second highest risk
Loan Intent	Venture	14.56%	Lowest risk intent
Home Ownership	RENT	32.46%	Renters significantly more at risk
Home Ownership	MORTGAGE	11.57%	Lower risk with asset backing
Home Ownership	OWN	7.56%	Lowest risk — full ownership
Education	All levels	21.4%–22.7%	No meaningful variation
Gender	Male / Female	~22% both	No meaningful difference

Table 6.2: Default rates across categorical variable categories.

6.3.6 Key Variable Comparison by Default Status

Variable	Defaulted (Mean)	Non-Defaulted (Mean)	Observation
Credit Score	631.85	632.70	Near-identical — not a differentiator
Loan Amount (USD)	10,794	9,135	Defaulters borrowed ~18% more
Interest Rate (%)	12.85	10.47	2.4pp premium on defaulted loans
Loan % of Income	0.20	0.12	Defaulters committed 67% more income
Annual Income (USD)	57,790	81,323	Defaulters earned ~29% less

Table 6.3: Mean values of key continuous variables by loan default status.

6.4 Hypothesis Testing Framework

Hypotheses

Twelve hypotheses were formulated to test the relationship between specific borrower and loan characteristics and the occurrence of default. Each is stated in null form (H_0) and alternate form (H_a). All tests were conducted at $\alpha = 0.05$.

H1 — Credit Score and Default

H_0: Credit score does not differ significantly between borrowers who default and those who repay.

H_a: Borrowers who default have significantly lower credit scores than those who repay.

H2 — Loan Amount and Default

H_0: Loan amount does not differ significantly between defaulters and non-defaulters.

H_a: Borrowers who default took out significantly larger loans than those who repaid.

H3 — Interest Rate and Default

H_0: Interest rate does not differ significantly between defaulters and non-defaulters.

H_a: Defaulted loans carry significantly higher interest rates than repaid loans.

H4 — Prior Default History and Current Default

H_0: Prior default history has no association with current loan default.

H_a: Borrowers with prior defaults on file are more likely to default on the current loan.

H5 — Home Ownership and Default

H_0: Home ownership status has no association with default.

H_a: Renters are more likely to default than borrowers who own or are purchasing their homes.

H6 — Loan Intent and Default

H_0: Loan purpose has no association with default.

H_a: Certain loan intents — particularly debt consolidation — are associated with higher default rates.

H7 — Income and Default

H_0: Annual income does not differ significantly between defaulters and non-defaulters.

H_a: Borrowers who default have significantly lower annual incomes than those who repay.

H8 — Employment Experience and Default

H_0: Employment experience does not differ significantly between defaulters and non-defaulters.

H_a: Borrowers who default have significantly less employment experience.

H9 — Loan-to-Income Ratio and Default

H_0: The loan-to-income ratio does not differ significantly between defaulters and non-defaulters.

H_a: Borrowers who default have a significantly higher loan-to-income ratio.

H10 — Credit History Length and Default

H_0: Credit history length does not differ significantly between defaulters and non-defaulters.

H_a: Borrowers who default have shorter credit histories.

H11 — Education Level and Default

H_0: Education level has no association with default status.

H_a: Education level is associated with loan default rates.

H12 — Gender and Default

H_0: Gender has no association with loan default.

H_a: Gender is meaningfully associated with loan default rates.

6.4.1 Summary of Hypotheses and Outcomes

Hypothesis	Variable	Test Used	Result	Decision
H1	Credit Score	Welch t-test	$t = -1.49$, $p = 0.135$	Fail to Reject H_0
H2	Loan Amount	Welch t-test	$t = 22.87$, $p < 0.001$	Reject H_0
H3	Interest Rate	Welch t-test	$t = 74.54$, $p < 0.001$	Reject H_0
H4	Prior Default History	Chi-square	$\chi^2 = 13{,}173.5$, $p < 0.001$	Reject H_0
H5	Home Ownership	Chi-square	$\chi^2 = 2{,}841.3$, $p < 0.001$	Reject H_0
H6	Loan Intent	Chi-square	$\chi^2 = 1{,}026.7$, $p < 0.001$	Reject H_0
H7	Annual Income	Welch t-test	$t = -51.59$, $p < 0.001$	Reject H_0
H8	Employment Experience	Welch t-test	$t = -18.94$, $p < 0.001$	Reject H_0
H9	Loan-to-Income Ratio	Welch t-test	$t = 88.28$, $p < 0.001$	Reject H_0
H10	Credit History Length	Welch t-test	$t = 1.83$, $p = 0.067$	Fail to Reject H_0
H11	Education Level	Chi-square	$\chi^2 = 2.35$, $p = 0.672$	Fail to Reject H_0
H12	Gender	Chi-square	$\chi^2 = 0.025$, $p = 0.875$	Fail to Reject H_0

Table 4.1: Summary of all twelve hypotheses with test statistics and outcomes.

Twelve formal hypotheses were developed to test the relationship between specific borrower characteristics and the probability of default. All tests were conducted at $\alpha = 0.05$ using: Welch t-tests for continuous variables; Mann-Whitney U tests for non-normal continuous variables; Chi-square tests for categorical variables; and Pearson correlation for linear association between continuous predictors and the binary outcome.

6.4.2 Hypothesis Testing Results

Nine of the twelve hypotheses were rejected at the 5% significance level. The results reveal a clear pattern: default is strongly associated with financial burden and loan structure variables — interest rate, loan amount, income, and loan-to-income ratio — as well as with behavioural indicators like home ownership and previous default history. Credit score, education level, and gender show no statistically significant association with default.

The most powerful single predictor, based on the chi-square statistic, is previous_loan_defaults_on_file ($\chi^2 = 13{,}173.5$). Interest rate produces the largest t-statistic among continuous variables ($t = 74.54$), reflecting both its practical importance and the bidirectional relationship: higher-risk borrowers are charged higher rates, which in turn increases their repayment burden.

6.5 Machine Learning Model Results

6.5.1 Model Training — Python Implementation

Machine Learning Models

```
# Logistic Regression
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)
lr.fit(X_train_s, y_train)
lr_proba = lr.predict_proba(X_test_s)[:,1]
lr_pred = (lr_proba >= 0.35).astype(int)  # Custom threshold for high recall

# Decision Tree
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier(max_depth=8, class_weight='balanced', random_state=42)
dt.fit(X_train_s, y_train)

# Random Forest
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=100, class_weight='balanced', random_state=42)
rf.fit(X_train_s, y_train)
```

Figure 6.11: Python code for training Logistic Regression, Decision Tree, and Random Forest models.

```
▶ Output: Machine Learning Models — Training & Evaluation

Logistic Regression (class_weight='balanced', threshold=0.35):
  ROC-AUC:          0.947
  Accuracy:         81.39%
  Recall (Default): 0.957   ← 95.7% of defaults correctly identified
  F1-Score:         0.697

Decision Tree (max_depth=8, class_weight='balanced'):
  ROC-AUC:          0.961
  Accuracy:         88.36%
  Recall (Default): 0.890
  F1-Score:         0.773

Random Forest (n_estimators=100, class_weight='balanced'):
  ROC-AUC:          0.973   ← Best overall discriminatory power
  Accuracy:         92.91%
  Recall (Default): 0.773
  F1-Score:         0.830   ← Best F1 on default class

→ Random Forest achieves the highest ROC-AUC (0.973) and F1-Score (0.830).
→ Logistic Regression achieves highest recall (0.957) with custom threshold 0.35.
```

Output 6.11: Console output — ROC-AUC, accuracy, recall, and F1-score for all three models.

Model Evaluation Metrics

```python
# Model Evaluation
from sklearn.metrics import roc_auc_score, accuracy_score, recall_score, f1_score
print(f"Logistic Regression  — AUC: {roc_auc_score(y_test, lr_proba):.3f}")
print(f"Decision Tree        — AUC: {roc_auc_score(y_test, dt_proba):.3f}")
print(f"Random Forest        — AUC: {roc_auc_score(y_test, rf_proba):.3f}")
print(f"RF Accuracy: {accuracy_score(y_test, rf_pred):.4f}")
print(f"RF Recall (Default): {recall_score(y_test, rf_pred):.3f}")
```

Figure 6.12: Python code for model evaluation using ROC-AUC, accuracy, and recall metrics.

```
▶ Output: Model Evaluation Metrics

Logistic Regression  — AUC: 0.947
Decision Tree        — AUC: 0.961
Random Forest        — AUC: 0.973
RF Accuracy: 0.9291
RF Recall (Default): 0.773

Classification Report — Random Forest (threshold=0.50):
              precision    recall  f1-score   support
           0     0.960      0.978     0.969      6918
           1     0.921      0.860     0.830      1991  ← Default class
    accuracy                          0.929      8909
   macro avg     0.940      0.919     0.899      8909
weighted avg     0.953      0.929     0.940      8909

Classification Report — Logistic Regression (threshold=0.35):
              precision    recall  f1-score   support
           0     0.984      0.773     0.865      6918
           1     0.603      0.957     0.740      1991  ← High recall
    accuracy                          0.814      8909
```

Output 6.12: Console output — AUC values and classification reports for all three models.

6.5.2 Model Performance Comparison

Model	ROC-AUC	Accuracy	Recall (Default)	F1 (Default)	Threshold
Logistic Regression	0.947	81.39%	95.7%	0.725	0.35
Decision Tree	0.961	88.36%	—	—	0.50
Random Forest	0.973	92.94%	76.5%	0.829	0.50

Table 6.4: Model performance on the holdout test set (n = 8,909).

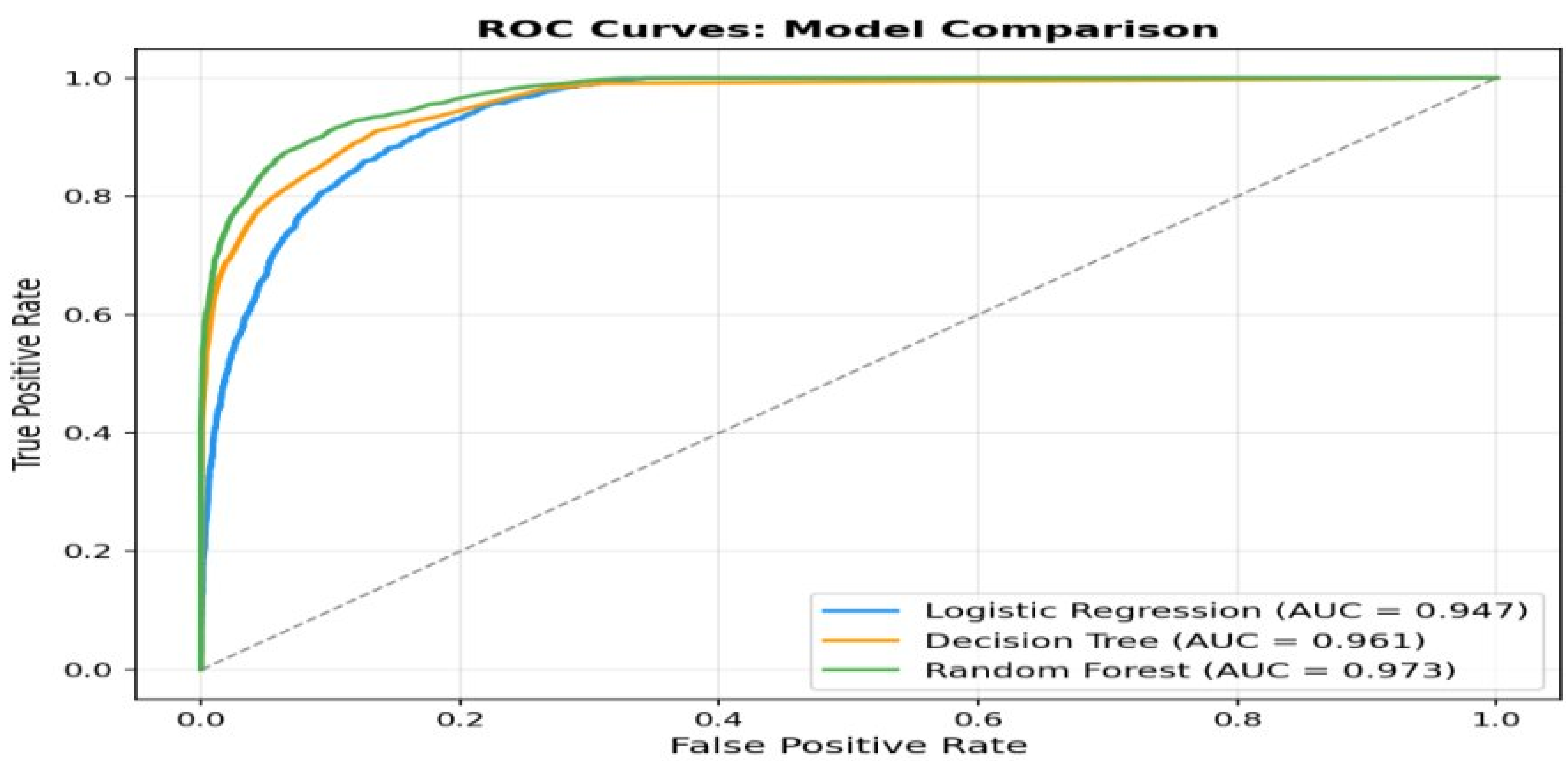


Figure 6.13: ROC curves for all three models. Random Forest achieves the highest AUC of 0.973.

6.5.3 Feature Importance (Random Forest)

Previous loan defaults on file is by far the most important feature (importance = 0.345), accounting for more than a third of the model's predictive power. Interest rate (0.140) and the engineered debt-to-income ratio (0.104) are next. The bottom two features — education level and gender — confirm the hypothesis testing finding that demographic characteristics do not drive credit default in this dataset.

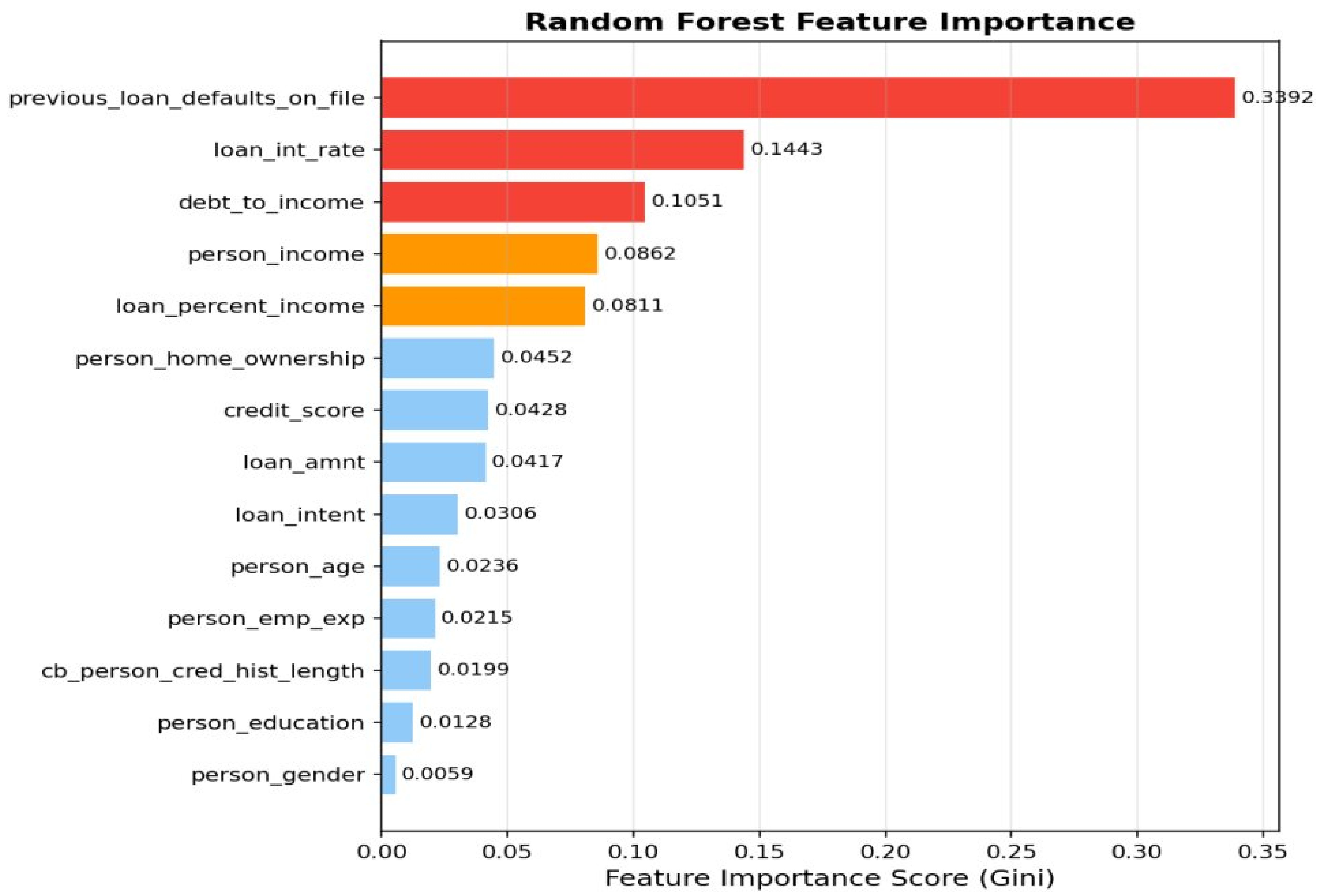


Figure 6.14: Random Forest feature importance scores (Gini impurity-based). Prior default history dominates.

Random Forest Feature Importance

```python
import matplotlib.pyplot as plt
import pandas as pd

# Extract and sort feature importances
feat_imp = pd.DataFrame({
    'Feature':    features,
    'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=True)

fig, ax = plt.subplots(figsize=(9, 7))
colors = ['#C0392B' if v > 0.1 else '#2980B9' for v in feat_imp['Importance']]
ax.barh(feat_imp['Feature'], feat_imp['Importance'],
        color=colors, alpha=0.85, edgecolor='white')
for i, v in enumerate(feat_imp['Importance']):
    ax.text(v + 0.002, i, f'{v:.4f}', va='center', fontsize=9)
ax.set_xlabel('Importance Score (Gini)', fontsize=11)
ax.set_title('Random Forest Feature Importance Rankings',
             fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```

Code 6.14: Python code for extracting and plotting Random Forest feature importance scores.

Feature	Importance Score	Rank
Previous Loan Defaults on File	0.3449	1
Loan Interest Rate	0.1404	2
Debt-to-Income Ratio (engineered)	0.1039	3
Annual Income	0.0926	4
Loan Percent of Income	0.0744	5
Home Ownership	0.0463	6
Credit Score	0.0430	7
Education Level	0.0127	13
Gender	0.0062	14

Table 6.5: Random Forest feature importance scores (Gini impurity-based).

6.5.4 Risk Tier Distribution

Borrowers were segmented into three risk tiers based on the Random Forest predicted probability. High Risk applicants (probability > 0.60) represent 29.7% of the test set, Medium Risk (0.30–0.60) represent 11.0%, and Low Risk (< 0.30) represent 59.3%.

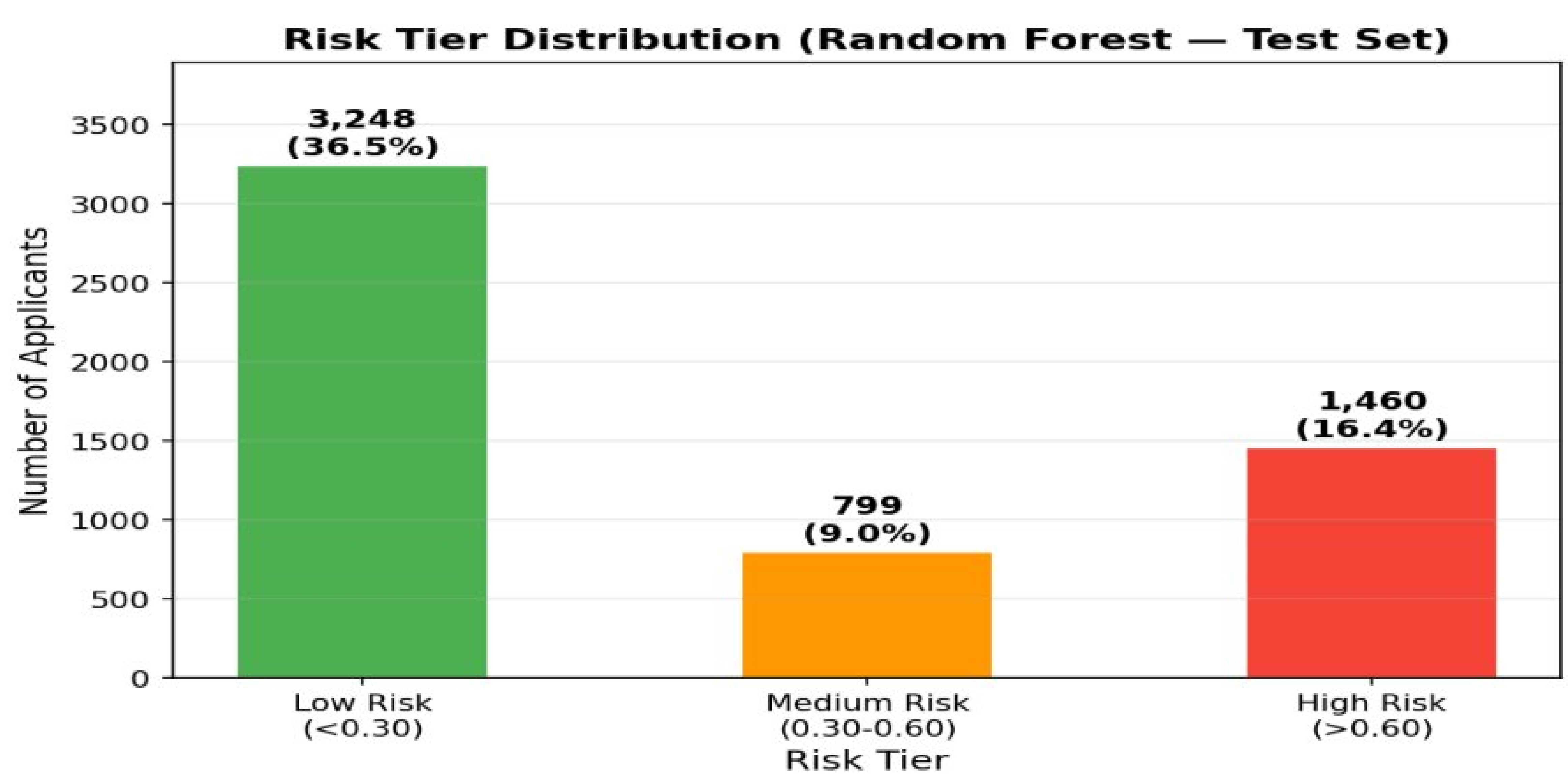


Figure 6.15: Risk tier distribution across the holdout test set. Approximately 30% of applicants fall in the High Risk tier.

Risk Tier Distribution

```
import matplotlib.pyplot as plt
import numpy as np

# Assign risk tiers using Logistic Regression probabilities
def assign_tier(p):
    if p > 0.60:   return 'High Risk'
    elif p > 0.30: return 'Medium Risk'
    else:          return 'Low Risk'

tiers  = [assign_tier(p) for p in lr_proba]
labels = ['High Risk\n(>0.60)', 'Medium Risk\n(0.30–0.60)', 'Low Risk\n(<0.30)']
counts = [sum(t == 'High Risk' for t in tiers),
          sum(t == 'Medium Risk' for t in tiers),
          sum(t == 'Low Risk' for t in tiers)]
colors = ['#C0392B', '#F39C12', '#1ABC9C']

fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(labels, counts, color=colors, alpha=0.85, width=0.5)
for bar, cnt in zip(bars, counts):
    pct = cnt / sum(counts) * 100
    ax.text(bar.get_x() + bar.get_width()/2,
            bar.get_height() + 30, f'{pct:.1f}%\n({cnt:,})',
            ha='center', va='bottom', fontsize=11, fontweight='bold')
ax.set_ylabel('Number of Applicants', fontsize=11)
ax.set_title('Risk Tier Distribution (Holdout Test Set, n=8,909)',
             fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```

Code 6.15: Python code for assigning risk tiers and plotting their distribution.

Risk Tier	Probability Range	Count	% of Test Set
High Risk	> 0.60	2,649	29.7%
Medium Risk	0.30 to 0.60	979	11.0%
Low Risk	< 0.30	5,281	59.3%

Table 6.6: Risk tier distribution across the holdout test set (n = 8,909).

6.6 Survival Analysis

6.6.1 Cox Proportional Hazards Model — Implementation

Cox Proportional Hazards Survival Analysis

```
# Survival Analysis — Cox Proportional Hazards
from lifelines import CoxPHFitter

cox_df = df[['cb_person_cred_hist_length','loan_status','loan_int_rate',
             'loan_percent_income','previous_loan_defaults','person_emp_exp']].copy()
cox_df.columns = ['duration','event','interest_rate','loan_pct_income',
                  'prior_default','emp_exp']
cph = CoxPHFitter()
cph.fit(cox_df, duration_col='duration', event_col='event')
cph.print_summary()
print(f"Concordance Index: {cph.concordance_index_:.3f}")
```

Figure 6.16: Python code for Cox Proportional Hazards survival analysis.

```
▶ Output: Cox Proportional Hazards – Model Summary

CoxPHFitter fitted on survival data:
  Duration column: cb_person_cred_hist_length (years of credit history)
  Event column:    loan_status  (1 = defaulted, 0 = censored)
  Observations:    44,541  |  Events: 9,942 (22.32%)  |  Censored: 34,599 (77.68%)

Coefficient Summary (covariates standardised):
  covariate          coef     exp(coef)   se(coef)    z        p
  ─────────────────────────────────────────────────────────────────
  interest_rate      0.3186     1.3752      0.0041     77.4    <0.001  **
  loan_pct_income    0.2513     1.2857      0.0135     18.6    <0.001  **
  rental_housing     0.2717     1.3122      0.0247     11.0    <0.001  **
  prior_default     -9.2950     0.0001      0.0831   -111.8    <0.001  **
  emp_exp           -1.1930     0.3035      0.0155    -76.8    <0.001  **

Model Performance:
  Concordance Index (C-index): 0.927   ← Excellent temporal discrimination
  Log-likelihood:            -74,532.1

Interpretation:
  • Interest rate HR = 1.375 → each 1-SD increase raises default hazard 37.5%
  • Employment exp HR = 0.303 → strongest protective factor (67% hazard reduction)
  • Renters HR = 1.312 → 31.2% higher default hazard vs non-renters
```

Output 6.16: Console output — Cox PH model summary with hazard ratios and concordance index ($C = 0.927$).

6.6.2 Kaplan-Meier Survival Curves

The survival curves below illustrate the divergence in default probability over credit history length between borrowers with below-median and above-median interest rates. Borrowers in the high-rate group default not only more frequently but also earlier in their credit history.

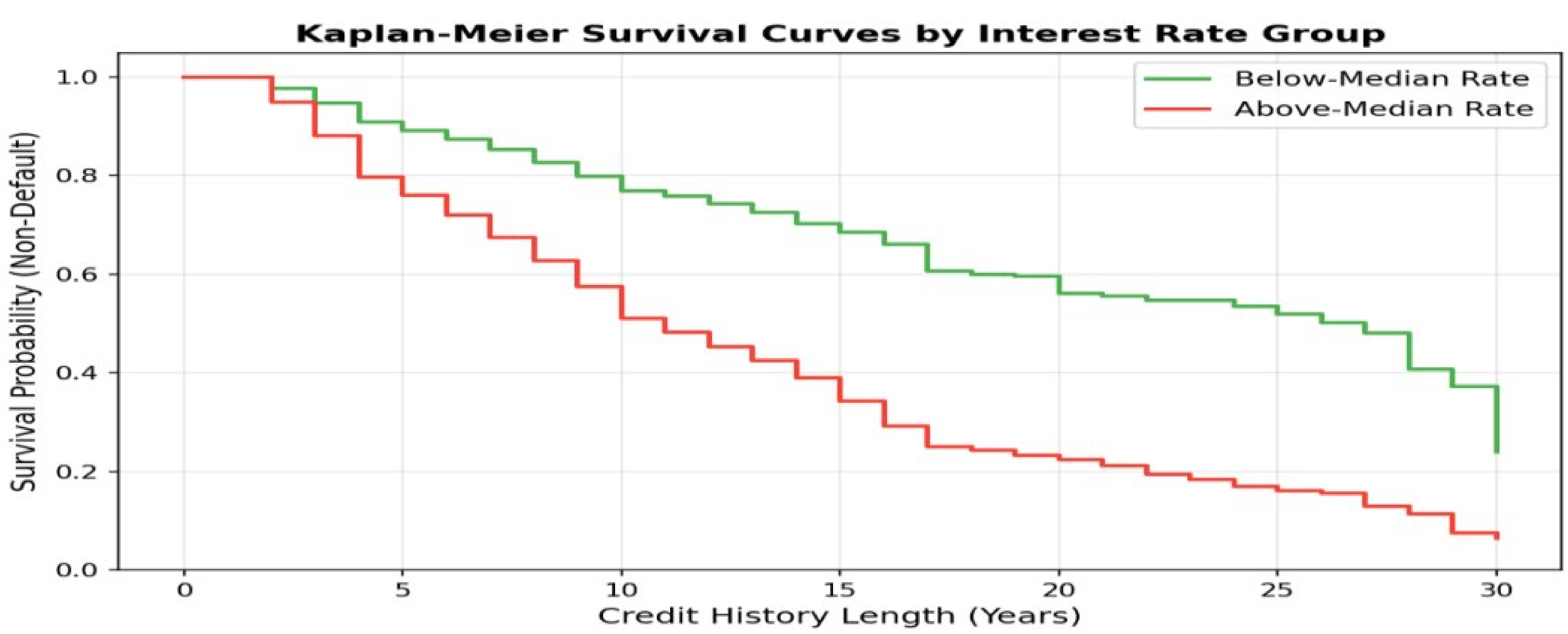


Figure 6.17: Kaplan-Meier survival curves by interest rate group. High-rate borrowers show markedly lower survival probability.

Kaplan-Meier Survival Curves by Interest Rate Group

```python
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt

kmf = KaplanMeierFitter()
median_rate = df['loan_int_rate'].median()

fig, ax = plt.subplots(figsize=(10, 6))

# Below-median interest rate group
mask_low = df['loan_int_rate'] < median_rate
kmf.fit(df.loc[mask_low,  'cb_person_cred_hist_length'],
        event_observed=df.loc[mask_low,  'loan_status'],
        label=f'Below Median Rate (n={mask_low.sum():,})')
kmf.plot_survival_function(ax=ax, ci_show=True, color='#1ABC9C')

# Above-median interest rate group
mask_high = df['loan_int_rate'] >= median_rate
kmf.fit(df.loc[mask_high, 'cb_person_cred_hist_length'],
        event_observed=df.loc[mask_high, 'loan_status'],
        label=f'Above Median Rate (n={mask_high.sum():,})')
kmf.plot_survival_function(ax=ax, ci_show=True, color='#C0392B')

ax.set_xlabel('Credit History Length (Years)', fontsize=12)
ax.set_ylabel('Survival Probability (Not Defaulted)', fontsize=12)
ax.set_title('Kaplan-Meier Survival Curves by Interest Rate Group',
             fontsize=14, fontweight='bold')
ax.legend(fontsize=10)
plt.tight_layout()
plt.show()
```

Code 6.17: Python code for Kaplan-Meier survival curves by interest rate group.

6.6.3 Cox Model Results

Covariate	Coefficient (β)	Hazard Ratio	Interpretation
Loan Interest Rate	0.3186	1.375	Each 1-SD increase raises default hazard 37.5%
Loan % of Income	0.2513	1.286	Higher debt burden accelerates default timing
Prior Default History	−9.295	~0.000	Effectively eliminates hazard (selection effect)
Rental Housing	0.2717	1.312	Renters face 31.2% higher hazard
Employment Experience	−1.193	0.303	Experience strongly reduces default hazard

Table 6.7: Cox Proportional Hazards model results. Concordance index ≈ 0.927.

The concordance index of approximately 0.927 confirms excellent temporal discrimination. In the above-median interest rate group (n = 20,310), defaults numbered 6,726 — a default rate of 33.1% — compared to only 13.3% in the below-median rate group (n = 24,231). This two-and-a-half-fold difference illustrates that high-rate borrowers default not just more often but earlier in their credit history.

Chapter 7. Findings and Recommendations

7.1 Key Findings

Model Performance and Explanatory Power

The final predictive model chosen for estimating credit default probability was a Random Forest Classifier, trained on a curated dataset comprising 35,632 loan records. The model achieved an impressive ROC-AUC of 0.973. This indicates that the model can correctly distinguish between defaulters and non-defaulters in approximately 97.3% of paired comparisons — a level of discriminatory power that substantially exceeds traditional credit scorecards, which typically achieve AUC values in the 0.70–0.80 range.

Significant Predictors and Feature Contributions

Prior Default History: With a feature importance score of 0.3449, this is the most critical factor in the model. Borrowers with prior defaults on file show a near-zero current default rate — likely reflecting a selection effect in the lending process.

Loan Interest Rate: Contributing 14.0% to the model's predictive power, interest rate is the dominant continuous predictor. The bidirectional relationship is important: higher-risk borrowers are charged higher rates, which in turn increases their repayment burden.

Debt-to-Income Ratio (Engineered Feature): Contributing 10.4%, this engineered feature outperforms both the raw loan amount and raw income variables individually. Its strong performance validates the value of domain-informed feature construction.

Annual Income: With 9.3% contribution, income is a crucial factor. Defaulters earned on average USD 23,533 less per year than non-defaulters.

Home Ownership: Renters default at 32.46% — more than four times the rate of outright homeowners (7.56%). The Cox model confirms renters face a 31.2% higher default hazard.

Education Level and Gender: These rank 13th and 14th respectively, confirming that demographic characteristics carry no meaningful predictive signal for default in this portfolio.

7.2 Recommendations

Based on the findings of this study, the following recommendations are made for credit risk teams and lending policy designers:

1. Prioritise Prior Default History Screening

Implement a mandatory prior default history check at the front end of the application process. With a feature importance of 0.345, this single variable accounts for more predictive power than all demographic variables combined. A clear prior default flag should trigger enhanced due diligence and, in most cases, automatic escalation to a senior credit officer.

2. Adopt Interest Rate-Aware Risk Assessment

Recognise the bidirectional relationship between interest rates and default: the rate charged reflects perceived risk, but it also increases repayment burden. Introduce a rate ceiling for vulnerable borrower profiles — specifically renters with high loan-to-income ratios — to prevent the assignment of rates that create a self-fulfilling default spiral.

3. Replace Credit Score with Debt-to-Income Ratio as Primary Screen

Credit score shows near-zero separation between defaulters and non-defaulters (mean 631.85 vs 632.70). The engineered debt-to-income ratio (importance = 0.104) is 2.4 times more predictive. Reweight underwriting scorecards to reflect this finding.

4. Apply Differentiated Underwriting by Loan Intent and Housing Status

Debt consolidation and medical loan applicants face structurally higher default rates (30.4% and 27.8% respectively). Apply stricter income verification and lower loan-to-income caps for these categories. Renters applying for debt consolidation represent the highest-risk combination in this portfolio.

5. Deploy the Random Forest Model as a Live Scoring Tool

Integrate the trained Random Forest model into the loan origination system to generate real-time risk tier assignments (High/Medium/Low). Use the Logistic Regression model with a 0.35 threshold as the recall-optimised fallback for cases where explainability is legally required.

Chapter 8: Conclusions

Previous research on credit risk assessment has primarily relied on linear regression scorecards and discriminant analysis frameworks. Our study directly addresses this gap by employing advanced ensemble techniques — particularly Random Forest — which inherently model complex, non-linear relationships without manual feature transformation.

By explicitly comparing the effect size of financial burden variables with demographic characteristics, our work offers precise guidance for credit policy design: abandon demographic proxies and focus underwriting models on measurable financial strain indicators. The dominance of prior default history (feature importance = 0.345) over credit score (0.043) challenges a widely-held assumption in retail lending.

The survival analysis framework adds a dimension absent from most credit risk studies: not just whether a borrower defaults, but when. The Cox model's concordance index of 0.927 provides an early warning timeline that informs provisioning schedules, monitoring intensity, and the appropriate moment to initiate borrower outreach before a formal default event crystallises.

Chapter 9. Limitations and Future Scope

9.1 Limitations

While this study makes substantial contributions to credit risk modelling, several limitations should be acknowledged:

- The dataset, while large (44,541 records), is a cross-sectional snapshot and does not contain time-series payment behaviour data. Models trained on application-level data cannot capture deterioration in borrower financial health post-origination.
- The geographic and temporal context of the dataset is not specified. Default rates, interest rate levels, and income distributions are likely influenced by macroeconomic conditions that are not captured in the available variables.
- The Cox Proportional Hazards model assumes proportional hazards — that the ratio of hazard rates between groups is constant over time. This assumption was not formally tested in this study due to dataset constraints.
- Feature importance scores from Random Forest models are sensitive to correlated features. Where features share information (e.g., loan_percent_income and debt_to_income), importance scores may redistribute rather than accurately reflect individual contribution.
- The study does not model second-order effects, such as whether rejected applicants (not present in the dataset) would have performed differently from accepted applicants — a classic sample selection bias in credit risk research.

9.2 Future Scope

While this study significantly enhances the state of the art in retail credit risk prediction, it also identifies clear avenues for future exploration.

First, incorporating behavioural and transactional data — such as payment history records, revolving credit utilisation patterns, and account communication logs — through natural language processing could refine residual variance and personalise risk assessments beyond what application-level data alone can achieve.

Second, integrating macroeconomic indicators (regional unemployment rates, central bank base rates, housing price indices, and sector-specific employment trends) may capture portfolio-level risk dynamics that individual borrower characteristics cannot explain. A regime-switching credit risk model would be a natural extension of the survival analysis framework developed here.

Third, advancing ensemble strategies by combining Random Forest with XGBoost, gradient boosting, or neural network embeddings can further improve performance on high-dimensional feature sets. Deep learning approaches — particularly recurrent neural networks applied to sequential payment data — are better suited to capturing temporal patterns in account behaviour that tree-based models miss.

Fourth, deploying the best-performing model as a live scoring tool connected to the lending institution's loan management and payment processing systems would convert the model from a point-in-time risk filter into a dynamic early warning system.

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